

The AI Monoculture Paradox for Financial Market

Working Paper — Mathematical Framework

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What this paper teaches us. Chenghan: This is the thing I want to deliver but not sure what position would be the best.

This paper offers three lessons for understanding AI adoption in financial markets. First, it qualifies the optimistic view that AI democratizes information. A standard argument is that broader access to more accurate predictive tools should improve market efficiency. Our analysis shows that this logic is incomplete. Even when an AI signal is more precise than human signals, its shared nature can reduce informational diversity: idiosyncratic human errors diversify away in aggregate order flow, whereas common AI errors survive aggregation, synchronize trades, and increase endogenous price impact. Thus, the relevant question is not only whether AI forecasts are more accurate, but also whether their errors are shared across market participants.

Second, the paper explains why rational market participants may endogenously create systemic correlation even when that correlation is welfare-reducing. The model does not assume correlated trading exogenously. Instead, traders choose whether to adopt a more precise but shared signal. Because each trader internalizes the private informational benefit of AI adoption but not the crowding externality imposed on others, decentralized adoption can generate excessive order-flow synchronization. In this sense, AI-induced correlation is not merely a technological feature of similar models or data sources; it is an equilibrium outcome of individually rational information choices.

Third, the model suggests a quantitative approach to AI governance inside financial institutions. A desk-level or portfolio-manager-level business case for AI adoption is not sufficient, because private profitability does not account for the firm-wide crowding and bias-amplification costs of shared signals. Full AI adoption should be evaluated against the firm-level objective $W^b(\alpha)$ and approved only when the objective is locally nondecreasing at $\alpha = 1$. More generally, institutions can internalize the externality by charging AI-driven strategies for the crowding cost they impose on the rest of the firm, thereby aligning desk-level adoption incentives with firm-level surplus. At the same time, the model does not imply that AI should be rejected merely because it is shared. Initial deployment can be beneficial when the baseline market is under-informative and the model's systematic bias is sufficiently small. The governance problem is therefore not a binary choice between using and banning AI, but a quantitative problem of deciding when to introduce AI, how far adoption should scale, and how much shared-signal exposure the institution should allow.

1 Introduction

The rapid proliferation of Artificial Intelligence (AI) in financial markets has democratized access to highly precise predictive analytics. Retail and institutional investors alike increasingly have access to shared generative AI tools, such as ChatGPT, when formulat-

ing trading strategies. While these AI advisors promise more accurate forecasts, emerging empirical evidence points to an unintended market-wide consequence: as traders substitute idiosyncratic analysis with a common AI signal, beliefs become increasingly aligned, leading to severe directional trade crowding. This raises a fundamental question: when a superior but shared AI advisor becomes readily accessible, will rational traders choose to adopt it, and if so, how will their collective adoption affect individual profits, market quality, and aggregate welfare?

Answering this question requires combining two forces that are usually studied separately. Classical market microstructure models and recent Mean-Field Game (MFG) analyses of trade crowding explain how correlated order flows generate endogenous liquidity costs through price impact. However, these frameworks typically take the signal structure as given: traders are endowed with private, public, or common signals rather than choosing whether to join a shared information technology. A separate literature on costly information acquisition does endogenize whether investors acquire information, but it does not study the distinctive tradeoff between private signal diversity and shared algorithmic precision under strategic trading and price impact. What is missing is therefore a model in which information choice itself is endogenous, and where the choice to adopt a shared signal feeds back into order-flow correlation, market depth, and welfare.

This paper provides such a framework by formalizing the “AI Monoculture Paradox,” a financial-market analogue of the Braess paradox. The central mechanism is simple but subtle. A shared AI signal can be privately attractive because it is more precise than an individual human signal. Yet when many traders adopt the same signal, they also inherit the same errors. The market therefore loses the diversification generated by idiosyncratic human mistakes, and individually rational adoption can generate collectively harmful crowding.

We begin with a two-stage finite-player model in which traders first choose their information source and then compete in a trading subgame. This minimal environment isolates the core strategic tension: a trader may prefer the AI signal because it gives a stronger private informational edge, but if many traders make the same choice, their beliefs and orders become synchronized. Thus AI adoption is not merely an information-acquisition decision; it is also a strategic commitment to a correlated trading technology.

We then develop a continuum mean-field formulation to study the same mechanism at market scale. The mean-field model is not merely a large-population extension; it also changes the informational burden of the adoption game. Rather than requiring each trader to know the exact signal choice of every other trader, the model asks each trader to optimize against a macroscopic adoption rate α , interpreted as the market-wide prevalence of AI usage. This is both more realistic in large financial markets, where traders may observe or infer aggregate AI usage but not the identities and information choices of all other participants, and more tractable, because the strategic environment

collapses from a high-dimensional adoption profile into a single aggregate state variable. This continuum formulation allows us to characterize equilibrium adoption, welfare, and planner benchmarks in a tractable aggregate environment.

Our first main result shows that when AI precision is moderately higher than human signal precision, individual rationality can create an adoption trap. Traders adopt the shared AI signal to preserve a private informational edge, but collective migration toward the same signal eliminates the diversification generated by idiosyncratic human errors. The common AI error survives aggregation, crowds order flow, reduces liquidity, and can make the decentralized AI-adoption equilibrium welfare-inferior to the pure-human benchmark.

We next study the regulatory implications of AI adoption by contrasting two planner benchmarks. A first-best planner, who controls both adoption rates and trading quantities, can exploit the operational efficiency of the AI signal while controlling aggregate exposure. As a result, the pure-human allocation is never first-best, and sufficiently precise AI leads to full first-best adoption. By contrast, an adoption-only second-best planner controls only the macroscopic adoption rate, while trading quantities remain decentralized. This restriction creates a regulatory divergence. In illiquid markets, the no-AI boundary becomes locally optimal for the second-best planner; under an additional monotone-crowding condition, this local prescription strengthens to a global prohibition, with $\alpha = 0$ as the unique constrained optimum. At the opposite boundary, full AI adoption becomes second-best optimal only after a strictly higher precision threshold than the decentralized full-adoption threshold, if such a second-best threshold exists. Thus, private incentives can lead traders to adopt a shared AI signal before full adoption is socially optimal, and only sufficiently high AI precision can restore the case for complete adoption under decentralized trading.

We then extend the analysis to heterogeneous private adoption costs. This extension rules out the possibility that the paradox is merely an artifact of homogeneous traders or corner adoption. When traders differ in their costs of accessing or implementing the AI signal, decentralized adoption is characterized by a fixed point between the private adoption gain and the marginal adoption cost. The equilibrium may therefore be interior rather than all-or-nothing. Nevertheless, the private-social wedge persists. The marginal adopter internalizes only the private utility gain from switching to the AI signal, but not the congestion externality imposed on other traders through greater synchronization of order flow. We show that, under economically interpretable right-tail conditions, the decentralized market over-adopts AI relative to the adoption-only second-best benchmark. Thus, even when adoption is gradual and cost-heterogeneous, rational AI adoption can remain excessive from the planner’s perspective.

Finally, we study systematic bias and model misspecification. Sections 2–4 give the AI signal a favorable benchmark: the AI error is common across adopters but structurally

unbiased. The resulting paradox is therefore a second-moment phenomenon: the common AI error does not shift average beliefs, but it changes the covariance structure of trades and survives aggregation in order flow. In Section 5, we introduce a first-moment distortion. The true AI signal contains a deterministic bias $b \neq 0$, while traders and market makers continue to behave under the subjective model $b = 0$. This separates subjective adoption incentives from objective informed-sector surplus. The decentralized adoption threshold is unchanged, because traders do not price the bias into their private adoption decisions. However, objective full-adoption surplus is strictly reduced by a quadratic bias penalty. Hence systematic misspecification weakly expands the set of precisions for which full AI adoption is welfare-dominated. For the adoption-only second-best planner, the bias lowers the marginal value of initial AI entry and can make full AI monoculture harder to justify at the boundary. Thus systematic bias does not replace the monoculture mechanism; it adds a first-moment blind spot on top of the original second-moment crowding externality.

Welfare criterion. Throughout the paper, our main welfare object is not a full social surplus measure. Instead, we study the aggregate surplus of the informed trading sector. This focus is natural in our Kyle-type environment: competitive market makers earn zero expected profits under the correctly specified semi-strong efficient benchmark, while noise-trader demand is modeled as an exogenous liquidity shock rather than as the solution to an endogenous welfare problem. Hence, the welfare comparisons below track the endogenous component of market surplus that is directly affected by AI adoption: informed traders' equilibrium trading profits.

We refer to this object as *gross informed-sector surplus*. When heterogeneous adoption costs are introduced in Section 4, we subtract the aggregate technology adoption cost and study *net informed-sector surplus*. In Section 5, where agents behave under a misspecified subjective model, we evaluate informed-sector surplus under the objective data-generating process while holding fixed the misspecified equilibrium trading and pricing rules.

1.1 Related Work

Our paper connects five strands of literature: strategic trading with correlated signals, endogenous information acquisition, mean-field games and trade crowding, Braess-style paradoxes in algorithmic systems, and AI-related model risk in financial markets.

Strategic trading with correlated and public signals. Our market environment builds on Kyle-type models of strategic trading under asymmetric information. [Kyle \(1985\)](#) studies a monopolistic informed trader trading against noise demand and competitive market makers. Subsequent work extends this framework to multiple informed

traders and correlated signals. [Holden and Subrahmanyam \(1992\)](#) and [Foster and Viswanathan \(1996\)](#), for example, show that correlation among informed traders' signals affects informed profits, market depth, and price informativeness. Related extensions study public or shared information in insider trading environments ([Luo, 2001](#); [Liu and Zhang, 2011](#); [Liu and Li, 2022](#)). These papers clarify how the correlation structure of information affects strategic trading, but the information structure is typically taken as exogenous. In contrast, our traders choose whether to rely on an idiosyncratic human signal or adopt a more precise but shared AI signal. The endogenous adoption margin is central to our monoculture mechanism.

Endogenous information acquisition and information choice. Our adoption game is related to the literature on costly information acquisition, but the economic object is different. [Grossman and Stiglitz \(1980\)](#) show that perfect informational efficiency is incompatible with costly information acquisition, because informed traders must be compensated for acquiring information. [Verrecchia \(1982\)](#) studies information acquisition in a noisy rational expectations economy where traders learn from both costly private inquiry and prices. [Veldkamp \(2006\)](#) shows that common information choices can generate excess comovement in asset prices.

These papers endogenize whether and what investors learn. We instead study whether traders adopt a shared information technology. The key tradeoff is not only information acquisition versus information cost, but private diversification versus shared precision. A human signal is less precise but idiosyncratic, so its noise diversifies away in the continuum. The AI signal is more precise but common, so its error survives aggregation and synchronizes order flow. This distinction makes the adoption rate itself a source of endogenous price impact and welfare loss. The heterogeneous-cost extension in Section 4 builds on the information-acquisition logic by allowing adoption to be gradual, but the social wedge comes from the crowding spillover of shared AI adoption rather than from information costs alone.

Mean-field games and trade crowding. Our continuum formulation is closest to mean-field models of trading with aggregate price impact. [Cardaliaguet and Lehalle \(2018\)](#) formulate optimal liquidation as a mean-field game of controls and show how correlated trading generates crowding costs through price impact. [Neuman and Vofsi \(2023\)](#) study a stochastic differential game in which agents respond to a common predictive signal under aggregate transient price impact. Related work includes [Casgrain and Jaimungal \(2020\)](#), [Lachapelle et al. \(2016\)](#), and [Fu et al. \(2021\)](#). These papers typically take the information structure or predictive signal as given. We instead make the market-wide adoption rate of a shared signal endogenous. This allows us to study how individually optimal information choices generate aggregate crowding and welfare losses.

Braess paradox and algorithmic monoculture. Our main mechanism is a financial-market analogue of a Braess paradox: an individually attractive technology can reduce aggregate surplus once adopted by many agents. [Acemoglu et al. \(2018\)](#) develop an informational Braess paradox in traffic networks, showing that additional information can worsen congestion. [Kleinberg and Raghavan \(2021\)](#) study algorithmic monoculture and show that a widely used algorithm can reduce welfare even when it is individually more accurate than independent human judgment. We extend this logic to strategic financial markets. Unlike route choice or screening applications, adoption is only the first stage in our model. Traders subsequently choose continuous trading quantities, respond to endogenous price impact, and affect the informativeness of prices through their collective order flow.

AI in financial markets, model risk, and regulation. Our paper also contributes to the emerging literature on AI in financial markets. Recent empirical work suggests that generative AI affects investor information processing and trading behavior ([Cheng et al., 2025](#); [Aghbabali et al., 2025](#)), while theoretical work studies AI-powered trading, algorithmic collusion, and price efficiency ([Dou et al., 2025](#)). Closely related empirical evidence is provided by [Leung et al. \(2025\)](#), who use generative AI to identify AI-driven equity open-end funds from fund prospectuses and show that AI usage is associated with higher trading correlation among funds employing similar AI-driven investment strategies. Their findings also suggest that the stability consequences of AI usage depend on the type of strategy: some AI strategies behave counter-cyclically in stressed markets, while sentiment-based strategies appear more pro-cyclical. These results provide an empirical counterpart to the synchronization channel in our model, while our analysis offers a tractable equilibrium mechanism through which privately attractive AI adoption can generate excessive crowding and welfare losses.

The regulatory part of our analysis connects to broader concerns about AI model risk and financial stability. Work on AI and systemic risk emphasizes that common data, common models, opacity, and strategic complementarities may amplify financial vulnerabilities ([Daníelsson et al., 2022](#)). Policy discussions similarly highlight that widespread AI adoption can create model-risk, governance, and financial-stability concerns. Our framework provides a tractable equilibrium channel for these concerns: a shared AI tool may be privately attractive, yet its widespread adoption can create crowding, excessive adoption, and amplified losses under systematic misspecification.

2 The Two-Trader Case: A Minimal Model of the AI Monoculture Paradox

We develop the core intuition through a minimal $N = 2$ model.

2.1 Market Environment

There are three periods and a single risky asset in the economy: period 0, at which the information (AI and human) is released and traders decide which signals they will use and announce it; period 1, at which trading takes place based on strategies of two sides; and period 2, at which the risky asset payoff is realized and consumption takes place.

The ex post liquidate value of the risky asset at period 1 is a random variable v , normally distributed with mean 0 and variance σ_v^2 . The assumption of a zero mean is a convenient normalization.

The economy is populated with three kinds of traders: two single risk neutral algorithmic traders, liquidity/noise traders, and competitive risk neutral market makers. The algorithmic traders have access to a shared AI signal and also have private information, and they submit orders based on the public information and their private information. Liquidity traders have an inelastic demand for the risky asset and their trading is exogenous. The presence of the liquidity traders not only serves as a camouflage and makes it impossible for the market makers to exactly infer the insider's private information, but also is the source of the profits to be exploited by the insider. The market makers observe the total orders and they set prices in a semi-strong efficient way.

The trading proceeds as follows. At period 0, all information has been announced. Traders can choose between two information strategies: relying on a private Human signal H , or adopting a shared AI signal A .

- **Human Signal (H):** Trader i observes a private, noisy signal $\theta_i = v + \eta_i$, where $\eta_i \sim \mathcal{N}(0, \sigma_H^2)$ are independent and identically distributed.
- **AI Signal (A):** Trader i observes a highly precise signal $s = v + \varepsilon_{AI}$, where $\varepsilon_{AI} \sim \mathcal{N}(0, \sigma_{AI}^2)$. s is accessed by all algorithmic traders.

The random variables v, η_i and ε_{AI} are mutually independent. Define $s_i \in \{H, A\}$ be the information structure trader i chooses. Let I_i denote the information set of trader i in Stage 2, where $I_i = \theta_i$ if $s_i = H$, and $I_i = s$ if $s_i = A$.

At period 1, each trader decides their trading quantity based on the information they received and the information structure their opponent chose. Let $q_i^{(s_1, s_2)}$ represent the equilibrium trading strategy for trader i in the specific subgame defined by the adoption profile (s_1, s_2) . In the meantime, the liquidity traders collectively submit an order ε_P , which is a normal random variable independent of v, η_i and ε_{AI} , with mean 0 and variance

σ_ε^2 . The total order of the insider and the liquidity traders is $\sum_{i=1}^2 q_i + \varepsilon_P$. After receiving the total order $\sum_{i=1}^2 q_i + \varepsilon_P$, the market makers take the opposite side of the incoming order and set the price $P \equiv P(\sum_{i=1}^2 q_i + \varepsilon_P)$ of the risky asset such that they expect to earn a zero profit conditional on the total order.

At period 2, the uncertainty is resolved and the risky asset payoff is realized. Trader i 's realized profit is given by:

$$\pi_i = q_i(v - P) - \frac{\phi}{2} q_i^2, \quad (1)$$

where $\phi > 0$ parameterizes trading costs and risk aversion (Almgren and Chriss, 2001).

2.2 Equilibrium

We aim to characterize the insiders' optimal trading strategies and the market makers' pricing rules. Since each trader makes sequential decisions across two stages—adopting an information structure and determining a trading quantity—we solve for the **Subgame Perfect Equilibrium (SPE)** of this two-stage game by first analyzing the trading subgame conditional on any adoption profile, and then evaluating the initial adoption game.

Definition 2.1 (Nash Equilibrium of the Trading Subgame). *Fix an adoption profile $(s_1, s_2) \in \{H, A\}^2$. A Nash equilibrium of the corresponding period-1 trading subgame consists of trading strategies*

$$q_1^{(s_1, s_2)}, \quad q_2^{(s_1, s_2)},$$

and a pricing rule $P^{(s_1, s_2)}(\cdot)$, such that:

1. **Mutual profit maximization.** *For each trader $i \in \{1, 2\}$, with $j \neq i$, the strategy $q_i^{(s_1, s_2)}$ is a best response to the opponent's equilibrium strategy $q_j^{(s_1, s_2)}$. That is,*

$$q_i^{(s_1, s_2)} \in \arg \max_{q_i \in L^2(I_i)} \mathbb{E} \left[\pi_i \left(q_i, q_j^{(s_1, s_2)}, P^{(s_1, s_2)} \right) \mid I_i, s_1, s_2 \right],$$

where $L^2(I_i)$ denotes the set of square-integrable trading rules measurable with respect to trader i 's signal I_i . The adoption profile (s_1, s_2) is fixed and publicly known within the subgame.

2. **Semi-strong market efficiency.** *Let*

$$y^{(s_1, s_2)} := q_1^{(s_1, s_2)} + q_2^{(s_1, s_2)} + \varepsilon_P.$$

Then the market maker's pricing rule satisfies

$$P^{(s_1, s_2)}(y^{(s_1, s_2)}) = \mathbb{E}[v \mid y^{(s_1, s_2)}].$$

We only consider linear Nash equilibria, which means the trading quantity is linear in the signal each trader receives.

Lemma 2.2 (Linear trading equilibrium). *Fix $\rho_H, \rho_A \in (0, 1)$. For every execution friction $\phi \geq 0$ and every adoption profile*

$$(s_1, s_2) \in \{H, A\}^2,$$

the corresponding period-1 trading subgame admits a unique linear Nash equilibrium. That is, there exist unique coefficients

$$c_{s_1, s_2}^1(\phi), \quad c_{s_1, s_2}^2(\phi),$$

and a unique market-depth coefficient

$$\kappa_{s_1, s_2}(\phi) > 0$$

such that

$$q_i^{(s_1, s_2)} = c_{s_1, s_2}^i(\phi) I_i, \quad P^{(s_1, s_2)}(y) = \kappa_{s_1, s_2}(\phi) y$$

form a linear Nash equilibrium of the trading subgame.

Moreover, the equilibrium objects

$$\kappa_{s_1, s_2}(\phi), \quad c_{s_1, s_2}^1(\phi), \quad c_{s_1, s_2}^2(\phi)$$

depend continuously on ϕ . At any value of ϕ for which the associated positive root is simple, this dependence is C^1 in a neighborhood of ϕ .

Proof. We look for linear strategies of the form

$$q_i^{(s_1, s_2)} = c_{s_1, s_2}^i I_i.$$

For a fixed adoption profile, write $c^1 = c_{s_1, s_2}^1$ and $c^2 = c_{s_1, s_2}^2$. Since the total order flow is

$$y = c^1 I_1 + c^2 I_2 + \varepsilon_P,$$

semi-strong market efficiency implies

$$P(y) = \mathbb{E}[v \mid y] = \kappa y,$$

where

$$\kappa = \frac{\text{Cov}(v, c^1 I_1 + c^2 I_2 + \varepsilon_P)}{\text{Var}(c^1 I_1 + c^2 I_2 + \varepsilon_P)}.$$

Because $\text{Cov}(v, I_i) = \sigma_v^2$ for both $I_i = \theta_i$ and $I_i = s$, this becomes

$$\kappa = \frac{(c^1 + c^2)\sigma_v^2}{\text{Var}(c^1 I_1 + c^2 I_2 + \varepsilon_P)}.$$

Given the opponent's linear strategy, trader i 's conditional objective is quadratic in q_i . The first-order condition is

$$\mathbb{E}[v \mid I_i] - \kappa \mathbb{E}[q_j \mid I_i, s_j] = (2\kappa + \phi)q_i. \quad (2)$$

The term $2\kappa q_i$ reflects trader i 's own price impact in the finite-player subgame.

Case 1: (H, H) . By symmetry,

$$c_{H,H}^1 = c_{H,H}^2 =: c_{H,H}.$$

Since

$$\mathbb{E}[q_2 \mid \theta_1, H] = c_{H,H} \rho_H \theta_1,$$

the first-order condition (2) gives

$$c_{H,H} = \frac{\rho_H}{(2 + \rho_H)\kappa_{H,H} + \phi}.$$

Substituting this expression into the pricing equation yields

$$\kappa_{H,H} = \frac{2c_{H,H}\sigma_v^2}{c_{H,H}^2(4\sigma_v^2 + 2\sigma_H^2) + \sigma_\varepsilon^2}.$$

Equivalently, $\kappa_{H,H}$ is a positive solution of

$$F_{H,H}(\kappa, \phi) := \kappa \sigma_\varepsilon^2 ((2 + \rho_H)\kappa + \phi)^2 - 2\rho_H \sigma_v^2 (\kappa + \phi) = 0.$$

This is a cubic polynomial in κ . Its coefficient sequence has exactly one sign change after zero coefficients are ignored. Hence, by Descartes' rule of signs, it has at most one positive root. Since

$$F_{H,H}(0, \phi) \leq 0, \quad F_{H,H}(\kappa, \phi) \rightarrow +\infty \quad \text{as } \kappa \rightarrow \infty,$$

and the positive root is explicit when $\phi = 0$, there exists exactly one positive root for every $\phi \geq 0$. This root determines a unique $c_{H,H}$, and hence a unique linear equilibrium in the (H, H) subgame.

Case 2: (A, A) . Again by symmetry,

$$c_{A,A}^1 = c_{A,A}^2 =: c_{A,A}.$$

Since both traders observe the same AI signal s ,

$$\mathbb{E}[q_2 \mid s, A] = c_{A,A}s.$$

Therefore (2) gives

$$c_{A,A} = \frac{\rho_A}{3\kappa_{A,A} + \phi}.$$

Substituting back into the pricing equation gives

$$\kappa_{A,A} = \frac{2c_{A,A}\sigma_v^2}{4c_{A,A}^2(\sigma_v^2 + \sigma_{AI}^2) + \sigma_\varepsilon^2}.$$

Equivalently, $\kappa_{A,A}$ solves

$$F_{A,A}(\kappa, \phi) := \kappa\sigma_\varepsilon^2(3\kappa + \phi)^2 - 2\rho_A\sigma_v^2(\kappa + \phi) = 0.$$

This cubic polynomial also has exactly one sign change in its coefficient sequence after zero coefficients are ignored. By the same Descartes argument as above, it has a unique positive root for every $\phi \geq 0$. This root uniquely determines $c_{A,A}$, and hence the (A, A) linear equilibrium is unique.

Case 3: (A, H) . Consider the asymmetric subgame in which trader 1 adopts the AI signal and trader 2 uses the human signal:

$$q_1 = c_{A,H}s, \quad q_2 = c_{H,A}\theta_2.$$

Using

$$\mathbb{E}[v \mid s] = \rho_A s, \quad \mathbb{E}[v \mid \theta_2] = \rho_H \theta_2,$$

and

$$\mathbb{E}[\theta_2 \mid s] = \rho_A s, \quad \mathbb{E}[s \mid \theta_2] = \rho_H \theta_2,$$

the two first-order conditions are

$$\begin{pmatrix} 2\kappa_{A,H} + \phi & \kappa_{A,H}\rho_A \\ \kappa_{A,H}\rho_H & 2\kappa_{A,H} + \phi \end{pmatrix} \begin{pmatrix} c_{A,H} \\ c_{H,A} \end{pmatrix} = \begin{pmatrix} \rho_A \\ \rho_H \end{pmatrix}.$$

Let

$$D(\kappa, \phi) := (2\kappa + \phi)^2 - \kappa^2\rho_A\rho_H.$$

Since $\rho_A, \rho_H \in (0, 1)$, we have

$$D(\kappa, \phi) \geq \kappa^2(4 - \rho_A \rho_H) > 0 \quad \text{for every } \kappa > 0, \phi \geq 0.$$

Thus the linear system is nonsingular, and

$$c_{A,H} = \frac{\rho_A(2\kappa + \phi - \kappa\rho_H)}{D(\kappa, \phi)}, \quad c_{H,A} = \frac{\rho_H(2\kappa + \phi - \kappa\rho_A)}{D(\kappa, \phi)}.$$

Substituting these expressions into the pricing equation

$$\kappa = \frac{(c_{A,H} + c_{H,A})\sigma_v^2}{\text{Var}(c_{A,H}s + c_{H,A}\theta_2 + \varepsilon_P)}$$

and clearing denominators yields a polynomial equation

$$F_{A,H}(\kappa, \phi) = 0.$$

For $\phi > 0$, this is a quintic polynomial in κ . For $\phi = 0$, after removing the trivial zero factor, the remaining equation has the same positive-root structure. A direct expansion shows that the relevant coefficient sequence has exactly one sign change for every

$$\rho_A, \rho_H \in (0, 1), \quad \phi \geq 0.$$

Therefore, by Descartes' rule of signs, the equation admits at most one positive root. On the other hand, the pricing residual is negative near $\kappa = 0$ and positive for sufficiently large κ , so a positive root exists. Hence the positive root is unique. Since $D(\kappa, \phi) > 0$, this unique positive root determines unique coefficients $c_{A,H}$ and $c_{H,A}$. Thus the (A, H) subgame admits a unique linear Nash equilibrium.

The (H, A) case is identical after relabeling the two traders, so it also admits a unique linear Nash equilibrium.

Dependence on ϕ . It remains to justify the stated parameter dependence. In each subgame, the market depth $\kappa_{s_1, s_2}(\phi)$ is characterized as the unique positive root of a polynomial equation

$$F_{s_1, s_2}(\kappa, \phi) = 0,$$

whose coefficients are continuous functions of ϕ . By the standard continuity of isolated polynomial roots, uniqueness of the positive root implies that

$$\phi \mapsto \kappa_{s_1, s_2}(\phi)$$

is continuous. Since the trading coefficients are explicit continuous functions of (κ, ϕ) , the maps

$$\phi \mapsto c_{s_1, s_2}^1(\phi), \quad \phi \mapsto c_{s_1, s_2}^2(\phi)$$

are also continuous.

Finally, if at some ϕ_0 the positive root is simple, i.e.

$$\partial_\kappa F_{s_1, s_2}(\kappa_{s_1, s_2}(\phi_0), \phi_0) \neq 0,$$

then the implicit function theorem implies that $\kappa_{s_1, s_2}(\phi)$ is C^1 in a neighborhood of ϕ_0 . Since the coefficients c_{s_1, s_2}^1 and c_{s_1, s_2}^2 are smooth functions of (κ, ϕ) wherever the denominator $D(\kappa, \phi)$ is nonzero, they inherit the same local C^1 regularity. This completes the proof. $\square$

2.3 Subgame Nash equilibrium

Definition 2.3 (Subgame Perfect Equilibrium of the Full Game). *Recall the definition of $q_1^{(s_1, s_2)}, q_2^{(s_1, s_2)}$ in Definition 2.1. For the two-stage market model beginning at Period 0, a **Subgame Perfect Equilibrium (SPE)** consists of an initial information adoption profile (s_1^*, s_2^*) , along with the subsequent trading quantities $q_1^{(s_1^*, s_2^*)}, q_2^{(s_1^*, s_2^*)}$ and the market maker's pricing rule P for every possible adoption profile, such that the initial adoption profile (s_1^*, s_2^*) maximizes each trader's ex-ante expected profit. Formally, for trader $i \in \{1, 2\}$ and any alternative signal strategy $s'_i \in \{H, A\}$:*

$$U_{s_i^*, s_j^*} > U_{s'_i, s_j^*} \quad (3)$$

where U_{s_1, s_2} is defined as the Ex-ante Expected Equilibrium Profit of Trader 1 when playing strategy s_1 against Trader 2 playing s_2 . This is defined as the unconditional expectation of the profit function, evaluated prior to the realization of the signals:

$$U_{s_1, s_2} = \mathbb{E} \left[\pi_1 \left(q_1^{(s_1, s_2)}, q_2^{(s_1, s_2)}, v, \varepsilon_P \right) \right], \quad (4)$$

where the expectation is taken over the joint distribution of the fundamental value v , the noise trader demand ε_P , and the signal noises $(\eta_1, \eta_2, \text{ or } \varepsilon_{AI})$.

Theorem 2.4 (The AI Monoculture Paradox). *Define equilibrium informed-sector surplus as the sum of expected profits*

$$W_{s_1, s_2} := U_{s_1, s_2} + U_{s_2, s_1}.$$

For any given human signal precision $\rho_H \in (0, 1)$, there exists an execution-friction threshold $\bar{\phi} > 0$ such that, for every $\phi \in [0, \bar{\phi})$, there exist a recovery threshold $\rho^*(\phi)$ and a

unique dominance threshold $\underline{\rho}_A^{Nash}(\phi)$ satisfying

$$\rho_H < \underline{\rho}_A^{Nash}(\phi) < \min\{\rho^*(\phi), 1\}.$$

For any AI signal precision

$$\rho_A \in \left(\underline{\rho}_A^{Nash}(\phi), \min\{\rho^*(\phi), 1\} \right),$$

the following hold:

1. Adopting the AI signal A is a strictly dominant strategy for both traders. Hence

$$(A, A, q_1^{(A,A)}, q_2^{(A,A)}, P^{(A,A)})$$

constitutes the subgame-perfect equilibrium of the two-stage game.

2. The resulting informed-sector surplus is strictly lower than under the pure-human benchmark:

$$W_{AA} < W_{HH}.$$

Proof. We first derive the closed form of U . Substituting FOC condition (2) back into the conditional expected profit simplifies the expression:

$$\mathbb{E}[\pi_i | I_i] = \frac{2\kappa + \phi}{2} q_i^2. \quad (5)$$

By the Law of Iterated Expectations, the unconditional expected profit is strictly proportional to the variance of the trader's optimal quantity:

$$\mathbb{E}[\pi_i] = \frac{2\kappa + \phi}{2} \mathbb{E}[q_i^2]. \quad (6)$$

So it suffices to calculate q_i which has expressions in Lemma 2.2. Note that the polynomial equation for κ has no closed solution when $\phi > 0$, so we structure the proof by first establishing the paradox for the baseline case without the exogenous penalty ($\phi = 0$), and then extending it to the general case ($\phi > 0$).

Part 1: Lower informed-sector surplus of Strategy A ($\phi = 0$)

It suffices to prove $U_{AA} < U_{HH}$. For the (A, A) subgame, setting $\phi = 0$ in the FOC yields $c_{A,A} = \frac{\rho_A}{3\kappa_{AA}}$. Substituting this into the market maker's pricing rule and solving for κ_{AA} yields:

$$\kappa_{AA} = \frac{\sqrt{2\rho_A}\sigma_v}{3\sigma_\varepsilon} \quad (7)$$

Substituting κ_{AA} and $c_{A,A}$ back into the profit function, the endogenous expected profit

under AI monoculture is:

$$U_{AA}(\rho_A) = \kappa_{AA} c_{AA}^2 (\sigma_v^2 + \sigma_{AI}^2) = \frac{\rho_A \sigma_v^2}{9\kappa_{AA}} = \frac{\sigma_v \sigma_\varepsilon \sqrt{\rho_A}}{3\sqrt{2}} \quad (8)$$

For the (H, H) subgame, setting $\phi = 0$ yields $c_H = \frac{\rho_H}{\kappa_{HH}(2+\rho_H)}$. Substituting this into the corresponding pricing rule yields:

$$\kappa_{HH} = \frac{\sqrt{2\rho_H} \sigma_v}{\sigma_\varepsilon(2 + \rho_H)} \quad (9)$$

The endogenous expected profit under diverse human signals is:

$$U_{HH}(\rho_H) = \kappa_{HH} c_{HH}^2 (\sigma_v^2 + \sigma_H^2) = \frac{\rho_H \sigma_v^2}{\kappa_{HH}(2 + \rho_H)^2} = \frac{\sigma_v \sigma_\varepsilon \sqrt{\rho_H}}{\sqrt{2}(2 + \rho_H)} \quad (10)$$

Consider a hypothetical scenario where the shared AI signal has the exact same precision as the private human signal ($\rho_A = \rho_H = \rho$). Comparing their endogenous profits:

$$U_{AA}(\rho) = \frac{\sigma_v \sigma_\varepsilon \sqrt{\rho}}{3\sqrt{2}}, \quad U_{HH}(\rho) = \frac{\sigma_v \sigma_\varepsilon \sqrt{\rho}}{\sqrt{2}(2 + \rho)}$$

Because $\rho \in (0, 1)$, we have $2 + \rho < 3$, which strictly implies $\frac{1}{3} < \frac{1}{2+\rho}$. Therefore, $U_{AA}(\rho) < U_{HH}(\rho)$.

On the other hand, because $U_{AA}(\rho_A)$ is strictly monotonically increasing and continuous in ρ_A , let us examine the limit as the AI signal becomes perfect ($\rho_A \rightarrow 1$):

$$U_{AA}(1) = \frac{\sigma_v \sigma_\varepsilon}{3\sqrt{2}}$$

We compare this against the baseline human profit $U_{HH}(\rho_H)$. The inequality $U_{AA}(1) > U_{HH}(\rho_H)$ holds if and only if:

$$\frac{1}{3} > \frac{\sqrt{\rho_H}}{2 + \rho_H} \iff 2 + \rho_H > 3\sqrt{\rho_H}$$

Letting $x = \sqrt{\rho_H} \in (0, 1)$, this becomes $x^2 - 3x + 2 > 0 \implies (x - 1)(x - 2) > 0$, which is strictly true for all $\rho_H \in (0, 1)$. Thus, a perfect AI signal ultimately overcomes the correlation penalty, yielding $U_{AA}(1) > U_{HH}(\rho_H)$.

Because $U_{AA}(\rho_H) < U_{HH}(\rho_H)$ and $U_{AA}(1) > U_{HH}(\rho_H)$, by the Intermediate Value Theorem, there exists a unique critical threshold $\rho^* \in (\rho_H, 1)$ such that $U_{AA}(\rho^*) = U_{HH}(\rho_H)$. For any AI signal precision in the interval $\rho_A \in (\rho_H, \rho^*)$, the AI signal is objectively more accurate than the human signal, yet the resulting informed-sector surplus under monoculture strictly decreases: $W_{AA} < W_{HH}$.

Part 2: Strict Dominance of Strategy A ($\phi = 0$)

To rigorously establish that (A, A) is the strictly dominant Nash equilibrium for $\rho_A \in (\underline{\rho}_A^{Nash}, \rho^*)$, It suffices to prove $U_{AH} > U_{HH}$ and $U_{AA} > U_{HA}$.

Solving the pricing rule $\kappa_{AH} = \frac{\text{Cov}(v, y)}{\text{Var}(y)}$ alongside the optimal asymmetric quantities c_{AH} and c_{HA} under the baseline $\phi = 0$ yields the exact closed-form endogenous expected profits:

$$U_{AH}(\rho_A) = \sigma_v \sigma_\varepsilon \frac{\rho_A(2 - \rho_H)^2}{(4 - \rho_A \rho_H) \sqrt{(\rho_A + \rho_H)(4 + \rho_A \rho_H) - 8\rho_A \rho_H}} \quad (11)$$

$$U_{HA}(\rho_A) = \sigma_v \sigma_\varepsilon \frac{\rho_H(2 - \rho_A)^2}{(4 - \rho_A \rho_H) \sqrt{(\rho_A + \rho_H)(4 + \rho_A \rho_H) - 8\rho_A \rho_H}} \quad (12)$$

Case 2a: Dominance against Strategy H. We must show $U_{AH}(\rho_A) > U_{HH}(\rho_H)$. Evaluating the endogenous profit U_{AH} at the boundary ($\rho_A = \rho_H$) yields:

$$\begin{aligned} U_{AH}(\rho_H) &= \sigma_v \sigma_\varepsilon \frac{\rho_H(2 - \rho_H)^2}{(4 - \rho_H^2) \sqrt{2\rho_H(4 + \rho_H^2) - 8\rho_H^2}} \\ &= \sigma_v \sigma_\varepsilon \frac{\rho_H(2 - \rho_H)^2}{(2 - \rho_H)(2 + \rho_H) \sqrt{2\rho_H(2 - \rho_H)^2}} \\ &= \frac{\sigma_v \sigma_\varepsilon \sqrt{\rho_H}}{\sqrt{2}(2 + \rho_H)} = U_{HH}(\rho_H) \end{aligned}$$

This confirms that when signals are equally precise and conditionally independent, the asymmetric subgame mathematically converges to the symmetric (H, H) subgame. Because $\frac{\partial U_{AH}}{\partial \rho_A} > 0$, it naturally follows that for any $\rho_A > \rho_H$, $U_{AH}(\rho_A) > U_{AH}(\rho_H) = U_{HH}(\rho_H)$. Thus, a trader strictly prefers to deviate to A when the opponent plays H .

Case 2b: Dominance against Strategy A. We must show $U_{AA}(\rho_A) > U_{HA}(\rho_A)$. Define the profit difference function $g(\rho_A) = U_{AA}(\rho_A) - U_{HA}(\rho_A)$ on the closed interval $[\rho_H, \rho^*]$.

At the lower endpoint ($\rho_A = \rho_H$): By the correlation penalty, $U_{AA}(\rho_H) < U_{HH}(\rho_H)$. Evaluating U_{HA} at $\rho_A = \rho_H$ gives $U_{HA}(\rho_H) = U_{HH}(\rho_H)$. Since $U_{AA}(\rho_H) < U_{HA}(\rho_H)$, we have $g(\rho_H) < 0$.

At the upper endpoint ($\rho_A = \rho^*$): By definition, $U_{AA}(\rho^*) = U_{HH}(\rho_H)$. The function $U_{HA}(\rho_A)$ is strictly monotonically decreasing with respect to ρ_A (the adverse price impact intensifies as the opponent becomes more precise). Because $\rho^* > \rho_H$, it strictly follows that $U_{HA}(\rho^*) < U_{HA}(\rho_H) = U_{HH}(\rho_H)$. Substituting $U_{HH}(\rho_H)$ for $U_{AA}(\rho^*)$, we obtain $U_{AA}(\rho^*) > U_{HA}(\rho^*)$. Therefore, $g(\rho^*) > 0$.

Because $g(\rho_A)$ is continuous, strictly monotonically increasing, $g(\rho_H) < 0$, and $g(\rho^*) > 0$, the Intermediate Value Theorem guarantees a unique root $\underline{\rho}_A^{Nash} \in (\rho_H, \rho^*)$ such that $g(\underline{\rho}_A^{Nash}) = 0$. For any $\rho_A \in (\underline{\rho}_A^{Nash}, \rho^*)$, $g(\rho_A) > 0$, meaning $U_{AA}(\rho_A) > U_{HA}(\rho_A)$. Consequently, (A, A) is the strictly dominant Nash equilibrium for the paradox interval.

Part 3: Extension to sufficiently small $\phi > 0$.

By Lemma 2.2, for each adoption profile $(s_1, s_2) \in \{H, A\}^2$, the linear trading equi-

librium is unique, and the corresponding equilibrium objects

$$\kappa_{s_1, s_2}(\phi), \quad c_{s_1, s_2}^1(\phi), \quad c_{s_1, s_2}^2(\phi)$$

depend continuously on ϕ . Therefore the four ex-ante equilibrium profit functions

$$U_{AA}(\rho_A, \phi), \quad U_{HH}(\rho_H, \phi), \quad U_{AH}(\rho_A, \phi), \quad U_{HA}(\rho_A, \phi)$$

are continuous in ϕ and in ρ_A on the relevant compact neighborhoods.

At $\phi = 0$, Parts 1 and 2 establish two strict inequalities on the nonempty interval

$$\left(\underline{\rho}_A^{Nash}(0), \rho^*(0) \right) \subset (\rho_H, 1) :$$

first,

$$U_{AA}(\rho_A, 0) < U_{HH}(\rho_H, 0),$$

and second,

$$U_{AH}(\rho_A, 0) > U_{HH}(\rho_H, 0), \quad U_{AA}(\rho_A, 0) > U_{HA}(\rho_A, 0).$$

The first inequality gives the surplus loss under AI monoculture, while the last two inequalities make A a strictly dominant signal choice.

Because these inequalities are strict and the payoff functions are continuous in ϕ , they persist under all sufficiently small perturbations of ϕ . Equivalently, there exists $\bar{\phi} > 0$ such that for every $\phi \in [0, \bar{\phi})$, the recovery threshold $\rho^*(\phi)$, defined by

$$U_{AA}(\rho^*(\phi), \phi) = U_{HH}(\rho_H, \phi),$$

and the dominance threshold $\underline{\rho}_A^{Nash}(\phi)$, defined by

$$U_{AA}(\underline{\rho}_A^{Nash}(\phi), \phi) = U_{HA}(\underline{\rho}_A^{Nash}(\phi), \phi),$$

are well-defined continuations of their frictionless counterparts and satisfy

$$\rho_H < \underline{\rho}_A^{Nash}(\phi) < \min\{\rho^*(\phi), 1\}.$$

Since the crossings at $\phi = 0$ are strict and the relevant payoff differences are continuous, shrinking $\bar{\phi}$ if necessary ensures that these thresholds remain well-defined. Therefore, for every

$$\rho_A \in \left(\underline{\rho}_A^{Nash}(\phi), \min\{\rho^*(\phi), 1\} \right),$$

the AI signal is strictly dominant for both traders, while the full-AI outcome satisfies

$$W_{AA}(\rho_A, \phi) < W_{HH}(\rho_H, \phi).$$

This proves the paradox for all sufficiently small $\phi > 0$. □

3 The Continuum Limit: A Mean-Field Game Formulation

To abstract away from the complex, two-stage strategic dependencies inherent in finite-player asymmetric subgames—where traders must explicitly condition their actions on the specific, realized choices of individual opponents—we extend the model to a continuum of informed traders. This Mean-Field Game formulation elegantly collapses the multi-stage interaction into a single-stage equilibrium, where individuals simply optimize against a macroscopic prior (α) representing the aggregate AI adoption rate. Furthermore, the continuum limit significantly enhances the analytical robustness of our framework regarding execution frictions. While the finite-player setting necessitates a frictionless baseline ($\phi \rightarrow 0$) to maintain algebraic tractability in asymmetric subgames, the price-taking nature of the MFG limit allows for a generalized characterization of the paradox across the entire parameter space of ϕ .

3.1 Setting and the Aggregation Problem

Assume a continuum of traders of measure one, indexed by $i \in [0, 1]$. Let $\alpha \in [0, 1]$ denote the macroscopic fraction of traders who adopt the shared AI signal ($s = v + \varepsilon_{AI}$), leaving a fraction $1 - \alpha$ relying on their private human signals ($\theta_i = v + \eta_i$). The random variables v , ε_{AI} , and the individual private noises η_i are mutually independent, and η_i are i.i.d. across i .

In the single-stage game, traders simultaneously choose their signal type and trading quantity. Suppose the macroscopic trading strategies are linear with respect to their signal. Throughout this section, c_A and c_H denote the mean-field strategy coefficients for AI and human adopters respectively; these are endogenous objects that depend on α and should be distinguished from the finite-player subgame coefficients c_{s_1, s_2}^i of Section 2.

By the exact Law of Large Numbers (LLN) for a continuum of independent random variables, **the idiosyncratic noises η_i of the human adopters perfectly cancel out in aggregation**. The total informed order flow Y_I thus simplifies deterministically conditional on v and ε_{AI} :

$$Y_I = \int_{i: s_i=A} c_A s \, di + \int_{i: s_i=H} c_H \theta_i \, di = \alpha c_A (v + \varepsilon_{AI}) + (1 - \alpha) c_H v \quad (13)$$

The competitive market makers observe the total order flow $y = Y_I + \varepsilon_P$, which is given by:

$$y = [\alpha c_A + (1 - \alpha) c_H] v + \alpha c_A \varepsilon_{AI} + \varepsilon_P \quad (14)$$

Because the market makers price the asset semi-strongly efficiently, $P = \mathbb{E}[v|y] = \kappa y$, where the endogenous market depth κ is determined by the projection theorem:

$$\kappa = \frac{\text{Cov}(v, y)}{\text{Var}(y)} = \frac{[\alpha c_A + (1 - \alpha) c_H] \sigma_v^2}{[\alpha c_A + (1 - \alpha) c_H]^2 \sigma_v^2 + \alpha^2 c_A^2 \sigma_{AI}^2 + \sigma_\varepsilon^2} \quad (15)$$

3.2 Equilibrium and Microscopic Strategies

We begin by formalising equilibrium in the continuum setting, in a manner consistent with Definition 2.3.

Definition 3.1 (Mean-field equilibrium). *Fix an aggregate adoption rate $\alpha \in [0, 1]$. Let*

$$q_i^H(\alpha) = c_H(\alpha) \theta_i, \quad q_i^A(\alpha) = c_A(\alpha) s$$

denote the corresponding optimal trading quantities, and let

$$P(\alpha) = \mathbb{E}[v \mid y(\alpha)]$$

be the semi-strong efficient pricing rule.

A mean-field equilibrium consists of an adoption rate $\alpha^ \in [0, 1]$, strategy coefficients $c_H(\alpha^*), c_A(\alpha^*)$, and a pricing rule $P(\alpha^*)$ such that:*

1. **Optimal trading.** *Given α^* , each trader who adopts signal type $m \in \{H, A\}$ solves the price-taking problem*

$$\max_{q_i \in L^2(I_i^m)} \mathbb{E} \left[q_i (v - P(\alpha^*)) - \frac{\phi}{2} q_i^2 \mid I_i^m \right],$$

and the optimizer is $q_i^m(\alpha^)$.*

2. **Optimal signal choice.** *Given the continuation values induced by α^* , no trader can profitably deviate in signal choice. Formally, define*

$$U_H(\alpha) \equiv \mathbb{E} \left[q_i^H(\alpha) (v - P(\alpha)) - \frac{\phi}{2} (q_i^H(\alpha))^2 \right], \quad U_A(\alpha) \equiv \mathbb{E} \left[q_i^A(\alpha) (v - P(\alpha)) - \frac{\phi}{2} (q_i^A(\alpha))^2 \right], \quad (16)$$

then

$$\alpha^* = 1 \implies U_A(1) \geq U_H(1),$$

$$\alpha^* = 0 \implies U_H(0) \geq U_A(0),$$

and

$$\alpha^* \in (0, 1) \implies U_A(\alpha^*) = U_H(\alpha^*).$$

3. **Market efficiency.** The price satisfies

$$P(\alpha^*) = \mathbb{E}[v \mid y(\alpha^*)].$$

Define the general **informed-sector surplus** for any arbitrary strategy profile (α, c_A, c_H) as $\mathcal{W}(\alpha, c_A, c_H)$. This object aggregates the ex-ante trading profits of the informed sector under the specified adoption rate and trading sensitivities. When $c_A(\alpha)$ and $c_H(\alpha)$ are determined endogenously by the decentralized trading equilibrium, we define the **equilibrium informed-sector surplus** at adoption rate α as

$$W(\alpha) \equiv \mathcal{W}(\alpha, c_A(\alpha), c_H(\alpha)) = \alpha U_A(\alpha) + (1 - \alpha) U_H(\alpha). \quad (17)$$

In particular,

$$W(1) = U_A(1), \quad W(0) = U_H(0),$$

so a surplus comparison between full AI adoption and pure human trading reduces to comparing $U_A(1)$ and $U_H(0)$ directly.

Explicit form of Utility under equilibrium strategy Because each individual trader i is of measure zero, their unilateral trading decision does not affect the macroscopic state $(\alpha, \kappa(\alpha), c_A(\alpha), c_H(\alpha))$. Taking the endogenous market depth $\kappa(\alpha)$ as exogenous from the perspective of trader i , trader i maximizes $\mathbb{E}[q_i(v - \kappa(\alpha)y) - \frac{\phi}{2}q_i^2 \mid I_i]$. The First-Order Condition yields $q_i = \frac{1}{\phi}\mathbb{E}[v - \kappa(\alpha)y \mid I_i]$. Let

$$Q(\alpha) \equiv \kappa(\alpha) [\alpha c_A(\alpha) + (1 - \alpha) c_H(\alpha)] \quad (18)$$

denote the fraction of the fundamental value v that is reflected in the equilibrium price at adoption rate α . Solving the FOC for each type of trader:

For human signal adopters ($I_i = \theta_i$):

$$\begin{aligned} \phi c_H(\alpha) &= \rho_H - \kappa(\alpha) [\alpha c_A(\alpha) \rho_H + (1 - \alpha) c_H(\alpha) \rho_H] = \rho_H (1 - Q(\alpha)) \\ c_H(\alpha) &= \frac{\rho_H}{\phi} (1 - Q(\alpha)) \end{aligned} \quad (19)$$

For AI signal adopters ($I_i = s$):

$$\begin{aligned} \phi c_A(\alpha) &= \rho_A (1 - Q(\alpha)) - \kappa(\alpha) \alpha c_A(\alpha) (1 - \rho_A) \\ c_A(\alpha) &= \frac{\rho_A}{\phi + \kappa(\alpha) \alpha (1 - \rho_A)} (1 - Q(\alpha)) \end{aligned} \quad (20)$$

We now formally define and derive the ex-ante expected profits. At adoption rate α , a human-signal trader submits $q_i^H(\alpha) = c_H(\alpha) \theta_i$ and an AI-signal trader submits $q_i^A(\alpha) = c_A(\alpha) s$, both facing equilibrium price $P(\alpha) = \kappa(\alpha) y$. as the ex-ante expected profit of a representative human-signal and AI-signal adopter who uses the best strategy they can have, respectively. These are the continuum analogues of equation (4); the dependence on α enters explicitly through $c_H(\alpha)$, $c_A(\alpha)$, and $\kappa(\alpha)$.

To derive a closed form, we apply an envelope argument analogous to equation (6). Since trader i takes $P(\alpha) = \kappa(\alpha) y$ as exogenous, the FOC gives $\mathbb{E}[v - P(\alpha) \mid I_i] = \phi q_i$. Taking the conditional expectation of the profit function (1):

$$\mathbb{E}[\pi_i \mid I_i] = q_i \mathbb{E}[v - P(\alpha) \mid I_i] - \frac{\phi}{2} q_i^2 = \phi q_i^2 - \frac{\phi}{2} q_i^2 = \frac{\phi}{2} q_i^2.$$

Applying the Law of Iterated Expectations and the linear strategies:

$$\begin{aligned} \mathbb{E}\left[\left(q_i^H(\alpha)\right)^2\right] &= c_H(\alpha)^2 \mathbb{E}[\theta_i^2] = c_H(\alpha)^2 (\sigma_v^2 + \sigma_H^2) = \frac{c_H(\alpha)^2 \sigma_v^2}{\rho_H}, \\ \mathbb{E}\left[\left(q_i^A(\alpha)\right)^2\right] &= c_A(\alpha)^2 \mathbb{E}[s^2] = c_A(\alpha)^2 (\sigma_v^2 + \sigma_{AI}^2) = \frac{c_A(\alpha)^2 \sigma_v^2}{\rho_A}. \end{aligned}$$

Therefore, the ex-ante expected profits are:

$$U_H(\alpha) = \frac{\phi}{2} c_H(\alpha)^2 \frac{\sigma_v^2}{\rho_H}, \quad U_A(\alpha) = \frac{\phi}{2} c_A(\alpha)^2 \frac{\sigma_v^2}{\rho_A}. \quad (21)$$

3.3 The AI Monoculture Paradox in the Continuum

We now establish the main paradox: there exists a parameter regime where AI monoculture is the unique stable equilibrium outcome, yet it strictly reduces informed-sector surplus relative to a world without AI.

Lemma 3.2 (Monotonicity of the Net Advantage on the Relevant Domain). *Let*

$$\mathcal{A}_+ := \{\alpha \in [0, 1] : \Delta(\alpha) > 0\}.$$

Then the relative advantage of the AI signal,

$$\Delta(\alpha) = U_A(\alpha) - U_H(\alpha),$$

is strictly decreasing on $\mathcal{A}_+$. Equivalently,

$$\Delta'(\alpha) < 0 \quad \forall \alpha \in \mathcal{A}_+ \cap (0, 1),$$

with the corresponding one-sided interpretation at the boundary points.

Proof. Let

$$x(\alpha) := \frac{\kappa(\alpha)}{\phi}, \quad A(\alpha) := 1 + \alpha(1 - \rho_A)x(\alpha),$$

and define

$$B(\alpha) := 1 + x(\alpha) \left[(1 - \alpha)\rho_H + \alpha \frac{\rho_A}{A(\alpha)} \right].$$

By construction,

$$B(\alpha) = \frac{1}{1 - Q(\alpha)}.$$

Using the equilibrium coefficients,

$$c_H(\alpha) = \frac{\rho_H}{\phi B(\alpha)}, \quad c_A(\alpha) = \frac{\rho_A}{\phi A(\alpha) B(\alpha)},$$

we obtain

$$U_H(\alpha) = \frac{\sigma_v^2 \rho_H}{2\phi B(\alpha)^2}, \quad U_A(\alpha) = \frac{\sigma_v^2 \rho_A}{2\phi A(\alpha)^2 B(\alpha)^2}.$$

Therefore

$$\Delta(\alpha) = \frac{\sigma_v^2}{2\phi} \frac{D(\alpha)}{B(\alpha)^2}, \quad D(\alpha) := \frac{\rho_A}{A(\alpha)^2} - \rho_H.$$

The pricing equation can be written compactly as

$$x(\alpha)B(\alpha)^2 = \lambda(\rho_H + \alpha D(\alpha)), \quad \lambda := \frac{\sigma_v^2}{\phi^2 \sigma_\varepsilon^2}.$$

Indeed,

$$\rho_H + \alpha D(\alpha) = (1 - \alpha)\rho_H + \alpha \frac{\rho_A}{A(\alpha)^2}.$$

It remains to show that $D(\alpha)/B(\alpha)^2$ is strictly decreasing whenever $D(\alpha) > 0$. Differentiating the identities defining $A(\alpha)$, $B(\alpha)$, and the pricing equation above, and then eliminating $x'(\alpha)$ and $B'(\alpha)$, gives the following expression. Suppressing the dependence on α for readability,

$$\Delta'(\alpha) = -\frac{\sigma_v^2 x}{\phi B^2} \frac{2A^3 D^2(\rho_H + \alpha D) + 2B\alpha\rho_A(1 - \rho_A)D + B\rho_A(1 - \rho_A)\rho_H + 2\rho_A(1 - \rho_A)\rho_H^2 x}{A^3 B(\rho_H + \alpha D) + 2A^3(\rho_H + \alpha D)^2 x + 2B\alpha^2 \rho_A(1 - \rho_A)x}.$$

Now take any $\alpha \in \mathcal{A}_+ \cap (0, 1)$. Since

$$\Delta(\alpha) = \frac{\sigma_v^2}{2\phi} \frac{D(\alpha)}{B(\alpha)^2},$$

the condition $\Delta(\alpha) > 0$ is equivalent to

$$D(\alpha) > 0.$$

Moreover,

$$x(\alpha) > 0, \quad A(\alpha) > 0, \quad B(\alpha) > 0, \quad \rho_H + \alpha D(\alpha) > 0.$$

The denominator in the displayed expression for $\Delta'(\alpha)$ is therefore strictly positive. The numerator is also strictly positive: its first term

$$2A^3D^2(\rho_H + \alpha D)$$

is strictly positive, and all remaining terms are nonnegative because $\rho_A \in (\rho_H, 1]$. Hence

$$\Delta'(\alpha) < 0.$$

This proves the claim. □

Theorem 3.3 (The Mean-Field AI Monoculture Paradox). *Assume $\rho_A > \rho_H$. Then there exists a unique adoption threshold $\underline{\rho}_A^{Nash} \in (\rho_H, 1)$ and a theoretical recovery root $\rho^* > \underline{\rho}_A^{Nash}$ such that, for any AI precision*

$$\rho_A \in (\underline{\rho}_A^{Nash}, \min\{\rho^*, 1\}),$$

the following hold:

1. *The game admits a unique Mean-Field Equilibrium characterized by complete AI monoculture ($\alpha^* = 1$).*
2. *The informed-sector surplus under this monoculture equilibrium is strictly lower than the pure human baseline:*

$$W(1) < W(0).$$

Proof. We divide the proof into four steps.

Step 1: Welfare comparison and the recovery root ρ^* . When $\alpha = 1$, substituting

$$c_A(1) = \frac{\rho_A}{\phi + \kappa_1}$$

into the pricing rule yields the implicit equation

$$\kappa_1(\phi + \kappa_1)^2 = \rho_A \phi \frac{\sigma_v^2}{\sigma_\varepsilon^2}. \tag{22}$$

Substituting this back into the payoff formula gives

$$U_A(1) = \frac{\phi \sigma_v^2 \rho_A}{2(\phi + \kappa_1)^2} = \frac{1}{2} \kappa_1 \sigma_\varepsilon^2.$$

Likewise, when $\alpha = 0$, the market depth is κ_0 and

$$U_H(0) = \frac{1}{2} \kappa_0 \sigma_\varepsilon^2.$$

Hence

$$W(1) < W(0) \iff U_A(1) < U_H(0) \iff \kappa_1 < \kappa_0.$$

Define

$$h(x) := \frac{x(\phi + x)^2 \sigma_\varepsilon^2}{\phi \sigma_v^2}.$$

Then (22) is precisely

$$h(\kappa_1) = \rho_A.$$

Since h is strictly increasing on $(0, \infty)$, the condition $\kappa_1 < \kappa_0$ is equivalent to

$$\rho_A < h(\kappa_0).$$

We therefore define

$$\rho^* := h(\kappa_0).$$

This is the recovery root: for every $\rho_A < \rho^*$, one has $W(1) < W(0)$, while at $\rho_A = \rho^*$, one has $W(1) = W(0)$.

Step 2: The full-adoption equilibrium condition and uniqueness of $\underline{\rho}_A^{Nash}$. At $\alpha = 1$, the no-deviation condition in Definition 3.1 reduces to

$$U_A(1) \geq U_H(1).$$

Introduce the full-adoption private gain

$$\Psi(\rho_A) := U_A(1; \rho_A) - U_H(1; \rho_A).$$

We first show that Ψ is strictly increasing in ρ_A , which will imply uniqueness of the decentralized full-adoption threshold.

Let

$$x_1 := \frac{\kappa_1}{\phi}.$$

Then (22) becomes

$$x_1(1 + x_1)^2 = \rho_A \frac{\sigma_v^2}{\phi^2 \sigma_\varepsilon^2}.$$

Differentiating implicitly yields

$$\frac{dx_1}{d\rho_A} = \frac{x_1(1+x_1)}{\rho_A(1+3x_1)}.$$

At $\alpha = 1$, equations (20) and (21) imply

$$U_A(1) = \frac{\sigma_v^2 \rho_A}{2\phi(1+x_1)^2}, \quad U_H(1) = \frac{\sigma_v^2 \rho_H (1 + (1 - \rho_A)x_1)^2}{2\phi(1+x_1)^2}.$$

Therefore

$$\Psi(\rho_A) = \frac{\sigma_v^2}{2\phi(1+x_1)^2} \left[\rho_A - \rho_H (1 + (1 - \rho_A)x_1)^2 \right].$$

A direct differentiation, using the formula for $dx_1/d\rho_A$, gives

$$\Psi'(\rho_A) = \frac{\sigma_v^2}{2\phi(1+x_1)^2(1+3x_1)} \left[1 + x_1 + 4\rho_H x_1 + (10 - 4\rho_A)\rho_H x_1^2 + 6(1 - \rho_A)\rho_H x_1^3 \right] > 0.$$

Hence Ψ is strictly increasing on $(\rho_H, 1]$.

To prove existence of a root, it suffices to show a sign change. It is convenient to use a sign-equivalent normalization of Ψ . Since

$$\frac{\sigma_v^2 \rho_H}{2\phi(1+x_1)^2} > 0,$$

we have

$$\Psi(\rho_A) \geq 0 \iff \sqrt{\frac{\rho_A}{\rho_H}} \geq 1 + (1 - \rho_A)x_1.$$

Define the normalized full-adoption gain

$$\widehat{\Psi}(\rho_A) := \sqrt{\frac{\rho_A}{\rho_H}} - 1 - (1 - \rho_A)x_1(\rho_A).$$

Then

$$\Psi(\rho_A) \geq 0 \iff \widehat{\Psi}(\rho_A) \geq 0.$$

At the lower endpoint,

$$\widehat{\Psi}(\rho_H) = -(1 - \rho_H)x_1(\rho_H) < 0,$$

and hence $\Psi(\rho_H) < 0$.

At the upper boundary of the paradox interval, if $\rho^* \leq 1$, then the same calculation as above gives

$$\widehat{\Psi}(\rho^*) > 0,$$

and therefore $\Psi(\rho^*) > 0$. If instead $\rho^* > 1$, then

$$\widehat{\Psi}(1) = \sqrt{\frac{1}{\rho_H}} - 1 > 0,$$

so $\Psi(1) > 0$.

Since Ψ is continuous and strictly increasing, there exists a unique root

$$\underline{\rho}_A^{Nash} \in (\rho_H, \min\{\rho^*, 1\})$$

such that

$$\Psi(\underline{\rho}_A^{Nash}) = 0.$$

Equivalently, for every $\rho_A > \underline{\rho}_A^{Nash}$, the full-adoption boundary $\alpha^* = 1$ satisfies $U_A(1) > U_H(1)$, and hence is an equilibrium candidate.

Step 3: Excluding other equilibria. We now show that, for every $\rho_A > \underline{\rho}_A^{Nash}$, the only Mean-Field Equilibrium is $\alpha^* = 1$.

First, $\alpha^* = 0$ cannot be an equilibrium whenever $\rho_A > \rho_H$. At $\alpha = 0$, equations (19)–(21) imply

$$c_H(0) = \frac{\rho_H}{\phi}(1 - Q(0)), \quad c_A(0) = \frac{\rho_A}{\phi}(1 - Q(0)).$$

Therefore

$$U_H(0) = \frac{\sigma_v^2 \rho_H}{2\phi}(1 - Q(0))^2, \quad U_A(0) = \frac{\sigma_v^2 \rho_A}{2\phi}(1 - Q(0))^2,$$

so

$$U_A(0) - U_H(0) = \frac{\sigma_v^2(1 - Q(0))^2}{2\phi}(\rho_A - \rho_H) > 0.$$

Hence the no-AI boundary $\alpha^* = 0$ is never an equilibrium.

Second, interior equilibria are impossible in the regime $\rho_A > \underline{\rho}_A^{Nash}$. By Step 2,

$$\Delta(1) = U_A(1) - U_H(1) = \Psi(\rho_A) > 0.$$

Suppose, toward a contradiction, that there exists an interior equilibrium $\alpha^* \in (0, 1)$. Then Definition 3.1 requires

$$\Delta(\alpha^*) = U_A(\alpha^*) - U_H(\alpha^*) = 0.$$

Since $\Delta(0) > 0$ and $\Delta(1) > 0$, if such an α^* existed there would be a largest point

$$\beta := \sup\{\alpha < 1 : \Delta(\alpha) \leq 0\} \in (0, 1),$$

with

$$\Delta(\beta) = 0, \quad \Delta(\alpha) > 0 \quad \forall \alpha \in (\beta, 1].$$

But then, on the interval $(\beta, 1] \subset \{\Delta > 0\}$, Lemma 3.2 implies that Δ is strictly decreasing. Therefore, for every $\alpha \in (\beta, 1)$,

$$\Delta(\alpha) > \Delta(1) > 0.$$

Letting $\alpha \downarrow \beta$ and using continuity yields

$$0 = \Delta(\beta) \geq \Delta(1) > 0,$$

a contradiction. Hence $\Delta(\alpha) > 0$ for all $\alpha \in [0, 1]$, and no interior equilibrium can satisfy the indifference condition $U_A(\alpha) = U_H(\alpha)$.

Combining the exclusion of $\alpha^* = 0$ and of interior equilibria, we conclude that for every

$$\rho_A \in (\underline{\rho}_A^{Nash}, \min\{\rho^*, 1\}),$$

the unique Mean-Field Equilibrium is $\alpha^* = 1$.

Step 4: The paradox interval and the unbreakable regime. By Step 1, for every

$$\rho_A < \rho^*,$$

we have

$$W(1) < W(0).$$

Therefore, on the interval

$$\rho_A \in (\underline{\rho}_A^{Nash}, \min\{\rho^*, 1\}),$$

the unique equilibrium is full AI monoculture, yet this equilibrium is welfare-inferior to the pure-human benchmark. This proves the paradox. $\square$

3.4 Planner for informed trading sector

We now study the normative implications of AI adoption through the informed-sector surplus criterion introduced above. The objective is to compare how different planner regimes affect the aggregate ex-ante trading surplus generated by informed traders. This reduced-form criterion is appropriate in our Kyle-type environment: competitive market makers earn zero expected profits by semi-strong efficiency, while noise-trader demand is exogenous.

We compare decentralized adoption with two planner problems under strictly positive execution frictions ($\phi > 0$). The distinction between the two problems is fundamental.

We use the terms *first-best* and *second-best* in their standard welfare-economic sense: a first-best benchmark is the allocation chosen by a planner who can directly control all relevant choice variables, whereas a second-best allocation is the best attainable allocation when some instruments are unavailable or some constraints must be respected.¹

An *unconstrained planner*, which we call the first-best planner, chooses both the macroscopic adoption rate and the aggregate trading exposures. This planner can exploit the higher precision of AI while directly controlling the amount of exposure placed on the shared signal. By contrast, an *adoption-only planner*, which we call the second-best planner, chooses only the adoption rate α , taking as given that trading quantities are subsequently chosen by decentralized traders.

This distinction generates the main regulatory divergence of the section. The first-best planner always finds it beneficial to introduce some AI exposure, and full AI adoption becomes first-best once the AI signal is sufficiently precise. The second-best planner faces a sharper constraint: because it cannot directly control trading intensities, changing α also changes the endogenous crowding externality created by decentralized order flow. As a result, the sign of the marginal value of AI entry at the no-AI boundary separates two market regimes. In an information-underextraction regime, marginal AI entry raises informed-sector surplus and the second-best optimum must involve positive AI adoption. In an information-overextraction regime, marginal AI entry lowers surplus at the boundary; under an additional monotone-crowding condition, this local force extends globally and the second-best planner uniquely chooses $\alpha^{SB} = 0$. Finally, at the opposite boundary, full AI adoption is delayed relative to decentralized incentives: if a global second-best full-adoption threshold exists, it must lie strictly above the decentralized full-adoption threshold $\underline{\rho}_A^{Nash}$.

Definition 3.4 (First-Best and Second-Best Planner Problems). *The planner problems maximize informed-sector surplus under two regulatory regimes.*

1. **Unconstrained Planner (First-Best).** *The planner chooses both the macroscopic AI adoption rate α and the microscopic trading sensitivities c_A and c_H . The objective is*

$$(\alpha^{FB}, c_A^{FB}, c_H^{FB}) \in \arg \max_{\alpha \in [0,1], c_A \geq 0, c_H \geq 0} \mathcal{W}(\alpha, c_A, c_H). \quad (23)$$

2. **Adoption-Only Planner (Second-Best).** *The planner chooses only the macroscopic adoption rate α , anticipating that traders subsequently select quantities according to the decentralized equilibrium derived above. The objective is*

$$\alpha^{SB} \in \arg \max_{\alpha \in [0,1]} W(\alpha). \quad (24)$$

¹This terminology follows the welfare-economics tradition of first-best and second-best analysis. See, for example, [Lipsey and Lancaster \(1956\)](#) for the classic theory of the second best and [Mas-Colell et al. \(1995, Ch. 10\)](#) for a standard general-equilibrium treatment of Pareto efficiency and welfare benchmarks.

When the maximizer is not unique, α^{SB} denotes any element of the argmax set.

Theorem 3.5 (First-Best Planner). *Consider the market with execution frictions $\phi > 0$ and AI precision $\rho_A \in (\rho_H, 1]$. In the unconstrained planner problem, the pure-human allocation is never first-best:*

$$\alpha^{FB} > 0.$$

Moreover, there exists a critical precision threshold

$$\rho_A^{FB} \in (\rho_H, 1)$$

such that for every $\rho_A \geq \rho_A^{FB}$, every first-best solution satisfies

$$\alpha^{FB} = 1.$$

Proof. Recall the aggregate informed order flow defined in (13). Writing

$$C_A := \alpha c_A, \quad C_H := (1 - \alpha) c_H, \quad C := C_A + C_H,$$

we have

$$Y_I = Cv + C_A \varepsilon_{AI}.$$

Hence the price impact coefficient is

$$\kappa = \frac{C \sigma_v^2}{C^2 \sigma_v^2 + C_A^2 \sigma_{AI}^2 + \sigma_\varepsilon^2},$$

and the aggregate trading revenue simplifies to

$$R(C_A, C_H) = \mathbb{E}[Y_I(v - P)] = \frac{C \sigma_v^2 \sigma_\varepsilon^2}{C^2 \sigma_v^2 + C_A^2 \sigma_{AI}^2 + \sigma_\varepsilon^2}.$$

The physical execution cost is

$$K(\alpha, C_A, C_H) = \frac{\phi \sigma_v^2}{2} \left[\frac{C_A^2}{\alpha \rho_A} + \frac{C_H^2}{(1 - \alpha) \rho_H} \right].$$

Minimizing over $\alpha \in (0, 1)$ for fixed (C_A, C_H) yields

$$K^*(C_A, C_H) = \frac{\phi \sigma_v^2}{2} \left(\frac{C_A}{\sqrt{\rho_A}} + \frac{C_H}{\sqrt{\rho_H}} \right)^2.$$

Hence the reduced first-best objective is

$$\mathcal{W}^*(C_A, C_H) = R(C_A, C_H) - K^*(C_A, C_H).$$

First, the zero-exposure allocation (i.e, $C_H = C_A = 0$) cannot be first-best. Indeed, for a pure-human exposure $C_H = C > 0$ and $C_A = 0$,

$$\mathcal{W}^*(0, C) = \frac{C \sigma_v^2 \sigma_\varepsilon^2}{C^2 \sigma_v^2 + \sigma_\varepsilon^2} - \frac{\phi \sigma_v^2 C^2}{2 \rho_H} > 0$$

for all sufficiently small $C > 0$, whereas $\mathcal{W}^*(0, 0) = 0$. Hence every first-best maximizer has strictly positive aggregate exposure.

We now show that a pure-human allocation cannot be first-best. Fix a total aggregate sensitivity $C > 0$, and consider the substitution path

$$(C_A, C_H) = (\varepsilon, C - \varepsilon), \quad \varepsilon \geq 0.$$

Along this path,

$$R(\varepsilon, C - \varepsilon) = \frac{C \sigma_v^2 \sigma_\varepsilon^2}{C^2 \sigma_v^2 + \varepsilon^2 \sigma_{AI}^2 + \sigma_\varepsilon^2},$$

so

$$\left. \frac{d}{d\varepsilon} R(\varepsilon, C - \varepsilon) \right|_{\varepsilon=0} = 0.$$

By contrast,

$$K^*(\varepsilon, C - \varepsilon) = \frac{\phi \sigma_v^2}{2} \left(\frac{\varepsilon}{\sqrt{\rho_A}} + \frac{C - \varepsilon}{\sqrt{\rho_H}} \right)^2,$$

hence

$$\left. \frac{d}{d\varepsilon} K^*(\varepsilon, C - \varepsilon) \right|_{\varepsilon=0} = \phi \sigma_v^2 \frac{C}{\sqrt{\rho_H}} \left(\frac{1}{\sqrt{\rho_A}} - \frac{1}{\sqrt{\rho_H}} \right) < 0,$$

because $\rho_A > \rho_H$. Therefore

$$\left. \frac{d}{d\varepsilon} \mathcal{W}^*(\varepsilon, C - \varepsilon) \right|_{\varepsilon=0} > 0.$$

Thus every first-best solution must involve positive AI exposure, which implies

$$\alpha^{FB} > 0.$$

We next prove that sufficiently precise AI leads the first-best planner to choose full AI adoption. Reparameterize the reduced objective by total sensitivity $C \geq 0$ and human exposure $H := C_H \in [0, C]$:

$$\widetilde{\mathcal{W}}(C, H) := \mathcal{W}^*(C - H, H) = \frac{C \sigma_v^2 \sigma_\varepsilon^2}{C^2 \sigma_v^2 + (C - H)^2 \sigma_{AI}^2 + \sigma_\varepsilon^2} - \frac{\phi \sigma_v^2}{2} \left(\frac{C - H}{\sqrt{\rho_A}} + \frac{H}{\sqrt{\rho_H}} \right)^2.$$

Since

$$R(C - H, H) \leq C \sigma_v^2, \quad K^*(C - H, H) \geq \frac{\phi \sigma_v^2}{2} C^2,$$

we obtain the global upper bound

$$\widetilde{\mathcal{W}}(C, H) \leq \sigma_v^2 \left(C - \frac{\phi}{2} C^2 \right).$$

Because the right-hand side is negative for $C > 2/\phi$ and $\widetilde{\mathcal{W}}(0, 0) = 0$, every first-best maximizer must satisfy

$$0 \leq H \leq C \leq \frac{2}{\phi}.$$

Hence it suffices to work on the compact set

$$\mathcal{D} := \left\{ (C, H) : 0 \leq H \leq C \leq \frac{2}{\phi} \right\}.$$

Differentiate $\widetilde{\mathcal{W}}$ with respect to H :

$$\frac{\partial \widetilde{\mathcal{W}}}{\partial H}(C, H) = \frac{2C\sigma_v^2\sigma_\varepsilon^2\sigma_{AI}^2(C-H)}{(C^2\sigma_v^2 + (C-H)^2\sigma_{AI}^2 + \sigma_\varepsilon^2)^2} - \phi\sigma_v^2 \left(\frac{C-H}{\sqrt{\rho_A}} + \frac{H}{\sqrt{\rho_H}} \right) \left(\frac{1}{\sqrt{\rho_H}} - \frac{1}{\sqrt{\rho_A}} \right).$$

For $(C, H) \in \mathcal{D}$, define

$$M := \sup_{0 \leq C \leq 2/\phi} \frac{2C\sigma_v^2\sigma_\varepsilon^2}{(C^2\sigma_v^2 + \sigma_\varepsilon^2)^2} < \infty.$$

Then

$$\frac{2C\sigma_v^2\sigma_\varepsilon^2\sigma_{AI}^2(C-H)}{(C^2\sigma_v^2 + (C-H)^2\sigma_{AI}^2 + \sigma_\varepsilon^2)^2} \leq M\sigma_{AI}^2 C.$$

On the other hand,

$$\left(\frac{C-H}{\sqrt{\rho_A}} + \frac{H}{\sqrt{\rho_H}} \right) \left(\frac{1}{\sqrt{\rho_H}} - \frac{1}{\sqrt{\rho_A}} \right) \geq \frac{C}{\sqrt{\rho_A}} \left(\frac{1}{\sqrt{\rho_H}} - \frac{1}{\sqrt{\rho_A}} \right).$$

Therefore

$$\frac{\partial \widetilde{\mathcal{W}}}{\partial H}(C, H) \leq C \left[M\sigma_{AI}^2 - \phi\sigma_v^2 \frac{1}{\sqrt{\rho_A}} \left(\frac{1}{\sqrt{\rho_H}} - \frac{1}{\sqrt{\rho_A}} \right) \right].$$

Now

$$\sigma_{AI}^2 = \sigma_v^2 \frac{1 - \rho_A}{\rho_A} \longrightarrow 0 \quad \text{as } \rho_A \uparrow 1,$$

whereas

$$\phi\sigma_v^2 \frac{1}{\sqrt{\rho_A}} \left(\frac{1}{\sqrt{\rho_H}} - \frac{1}{\sqrt{\rho_A}} \right) \longrightarrow \phi\sigma_v^2 \left(\frac{1}{\sqrt{\rho_H}} - 1 \right) > 0.$$

Hence there exists $\rho_A^{FB} < 1$ such that for all $\rho_A \geq \rho_A^{FB}$,

$$M\sigma_{AI}^2 < \phi\sigma_v^2 \frac{1}{\sqrt{\rho_A}} \left(\frac{1}{\sqrt{\rho_H}} - \frac{1}{\sqrt{\rho_A}} \right).$$

For such ρ_A , we have

$$\frac{\partial \widetilde{\mathcal{W}}}{\partial H}(C, H) < 0 \quad \forall (C, H) \in \mathcal{D} \text{ with } C > 0.$$

Therefore, for each fixed C , the map $H \mapsto \widetilde{\mathcal{W}}(C, H)$ is strictly decreasing on $[0, C]$, so its maximum is attained at $H = 0$. Thus every global maximizer of the reduced first-best problem satisfies $C_H = 0$.

Finally, for $C > 0$ and $H = 0$,

$$\widetilde{\mathcal{W}}(C, 0) = \frac{C \sigma_v^2 \sigma_\varepsilon^2}{C^2(\sigma_v^2 + \sigma_{AI}^2) + \sigma_\varepsilon^2} - \frac{\phi \sigma_v^2 C^2}{2 \rho_A} > 0$$

for all sufficiently small $C > 0$, whereas $\widetilde{\mathcal{W}}(0, 0) = 0$. Hence every first-best maximizer has $C > 0$. Given $C_H = 0$ and $C_A = C > 0$, the original cost

$$K(\alpha, C_A, 0) = \frac{\phi \sigma_v^2 C_A^2}{2 \alpha \rho_A}$$

is minimized at $\alpha = 1$. Therefore every first-best solution satisfies

$$\alpha^{FB} = 1.$$

□

Theorem 3.6 (Second-Best Planner). *Consider the constrained planner problem*

$$\alpha^{SB} \in \arg \max_{\alpha \in [0,1]} W(\alpha), \quad W(\alpha) = \alpha U_A(\alpha) + (1 - \alpha) U_H(\alpha).$$

Boundary derivatives are understood as one-sided derivatives. Define

$$\Delta(\alpha) := U_A(\alpha) - U_H(\alpha), \quad E(\alpha) := \alpha U'_A(\alpha) + (1 - \alpha) U'_H(\alpha),$$

so that, on the interior,

$$W'(\alpha) = \Delta(\alpha) + E(\alpha).$$

Then the constrained planner admits the following boundary and global characterizations.

1. **Use AI for Under-informative market.** *The right derivative at $\alpha = 0$ satisfies*

$$W'_+(0) = U_H(0) \frac{\rho_A - \rho_H}{\rho_H} \left(\frac{1 - 2Q(0)}{1 + 2Q(0)} \right).$$

Hence

$$W'_+(0) > 0 \iff \rho_H < 2\phi \frac{\sigma_\varepsilon}{\sigma_v}, \quad W'_+(0) < 0 \iff \rho_H > 2\phi \frac{\sigma_\varepsilon}{\sigma_v}.$$

In particular, if

$$\rho_H < 2\phi \frac{\sigma_\varepsilon}{\sigma_v},$$

then every constrained optimum satisfies

$$\alpha^{SB} > 0.$$

2. **Global AI prohibition for over-informative market.** Suppose

$$\rho_H > 2\phi \frac{\sigma_\varepsilon}{\sigma_v},$$

so that $W'_+(0) < 0$. If, in addition, the endogenous crowding component is weakly decreasing:

$$E(\alpha) \leq E(0) \quad \forall \alpha \in [0, 1],$$

or equivalently, under differentiability,

$$E'(\alpha) \leq 0 \quad \forall \alpha \in (0, 1),$$

then W is strictly decreasing on $[0, 1]$. Consequently,

$$\alpha^{SB} = 0$$

is the unique global maximizer of the constrained planner's problem.

3. **Full adoption is delayed comparing to decentralized equilibrium.** Define

$$\mathcal{S}_{SB} := \left\{ \rho \in (\rho_H, 1] : 1 \in \arg \max_{\alpha \in [0, 1]} W(\alpha; \rho) \right\}.$$

If $\mathcal{S}_{SB} \neq \emptyset$, define the global full-adoption threshold

$$\rho_A^{SB} := \inf \mathcal{S}_{SB}.$$

Then

$$\rho_A^{SB} > \underline{\rho}_A^{Nash}.$$

If $\mathcal{S}_{SB} = \emptyset$, then complete AI adoption is never globally second-best optimal on the feasible precision domain.

Proof. We prove the three claims in order.

Use AI for Under-informative market. The derivative at the origin is

$$W'_+(0) = \frac{\sigma_v^2}{2\phi}(1 - Q(0))^2(\rho_A - \rho_H) - \frac{\sigma_v^2 \rho_H}{\phi}(1 - Q(0)) \frac{\partial Q}{\partial \alpha} \Big|_{\alpha=0},$$

which simplifies to

$$W'_+(0) = U_H(0) \frac{\rho_A - \rho_H}{\rho_H} \left(\frac{1 - 2Q(0)}{1 + 2Q(0)} \right).$$

This yields

$$W'_+(0) > 0 \iff \rho_H < 2\phi \frac{\sigma_\varepsilon}{\sigma_v}, \quad W'_+(0) < 0 \iff \rho_H > 2\phi \frac{\sigma_\varepsilon}{\sigma_v}.$$

If $W'_+(0) > 0$, then $\alpha = 0$ cannot maximize W on $[0, 1]$, so every constrained optimum satisfies

$$\alpha^{SB} > 0.$$

If $W'_+(0) < 0$, continuity implies that $W(\alpha) < W(0)$ for all sufficiently small $\alpha > 0$, so $\alpha = 0$ is a strict local maximizer.

Global AI prohibition for over-informative market. We now prove the global prohibition statement. By definition,

$$\Delta(\alpha) := U_A(\alpha) - U_H(\alpha), \quad E(\alpha) := \alpha U'_A(\alpha) + (1 - \alpha) U'_H(\alpha),$$

and therefore

$$W'(\alpha) = \Delta(\alpha) + E(\alpha) \quad \text{for } \alpha \in (0, 1).$$

Since $\rho_A > \rho_H$, we have

$$\Delta(0) > 0.$$

By Lemma 3.2, Δ is strictly decreasing on

$$A_+ := \{\alpha \in [0, 1] : \Delta(\alpha) > 0\}.$$

We claim that

$$\Delta(\alpha) \leq \Delta(0) \quad \forall \alpha \in [0, 1].$$

Indeed, suppose by contradiction that $\Delta(a) > \Delta(0)$ for some $a \in (0, 1]$. Since $\Delta(0) > 0$, define

$$b := \sup\{t \in [0, a] : \Delta(t) = \Delta(0)\}.$$

Then $b < a$, $\Delta(\alpha) > \Delta(0) > 0$ for all $\alpha \in (b, a]$, and therefore $(b, a] \subset A_+$. By the mean

value theorem applied on $[b, a]$, there exists $\xi \in (b, a)$ such that

$$\Delta'(\xi) = \frac{\Delta(a) - \Delta(b)}{a - b} > 0,$$

contradicting the strict decrease of Δ on A_+ . Hence the claim holds.

Now since E is weakly decreasing on $[0, 1]$, for every $\alpha \in (0, 1)$,

$$W'(\alpha) = \Delta(\alpha) + E(\alpha) \leq \Delta(0) + E(0) = W'_+(0).$$

Under the illiquidity condition

$$\rho_H > 2\phi \frac{\sigma_\varepsilon}{\sigma_v},$$

we have $W'_+(0) < 0$. Hence

$$W'(\alpha) < 0 \quad \forall \alpha \in (0, 1).$$

Thus W is strictly decreasing on $[0, 1]$, and $\alpha = 0$ is the unique global maximizer of the constrained planner's problem.

The full-adoption boundary. At the right boundary,

$$W'_-(1) = U_A(1) - U_H(1) + \frac{\partial U_A(\alpha)}{\partial \alpha} \Big|_{\alpha=1}.$$

Let

$$x_1 := \frac{\kappa_1}{\phi},$$

where, by (22), x_1 is the unique positive solution to

$$x_1(1 + x_1)^2 = \rho_A \frac{\sigma_v^2}{\phi^2 \sigma_\varepsilon^2}.$$

At $\alpha = 1$,

$$U_A(1) = \frac{\sigma_v^2 \rho_A}{2\phi(1 + x_1)^2}, \quad U_H(1) = \frac{\sigma_v^2 \rho_H (1 + (1 - \rho_A)x_1)^2}{2\phi(1 + x_1)^2}.$$

Hence the Nash full-adoption threshold $\underline{\rho}_A^{Nash}$ is characterized by

$$U_A(1) \Big|_{\rho_A = \underline{\rho}_A^{Nash}} = U_H(1) \Big|_{\rho_A = \underline{\rho}_A^{Nash}}.$$

Differentiating the equilibrium system at $\alpha = 1$ and evaluating at $\rho_A = \underline{\rho}_A^{Nash}$ gives

$$\frac{\partial U_A(\alpha)}{\partial \alpha} \Big|_{\alpha=1, \rho_A = \underline{\rho}_A^{Nash}} = - \frac{\sigma_v^2 \underline{\rho}_A^{Nash} (1 - \underline{\rho}_A^{Nash}) x_1}{\phi(1 + x_1)(1 + 3x_1)(1 + (1 - \underline{\rho}_A^{Nash})x_1)} < 0.$$

Therefore,

$$W'_-(1) \Big|_{\rho_A = \underline{\rho}_A^{Nash}} = \left(U_A(1) - U_H(1) \right) \Big|_{\rho_A = \underline{\rho}_A^{Nash}} + \frac{\partial U_A(\alpha)}{\partial \alpha} \Big|_{\alpha=1, \rho_A = \underline{\rho}_A^{Nash}} < 0.$$

Now define

$$\mathcal{S}_{SB} := \left\{ \rho \in (\rho_H, 1] : 1 \in \arg \max_{\alpha \in [0,1]} W(\alpha; \rho) \right\}.$$

If $\mathcal{S}_{SB} = \emptyset$, then complete AI adoption is never globally second-best optimal on the feasible precision domain, and there is nothing further to prove. Suppose instead that $\mathcal{S}_{SB} \neq \emptyset$, and define

$$\rho_A^{SB} := \inf \mathcal{S}_{SB}.$$

Since $W'_-(1; \rho)$ is continuous in ρ , the strict inequality

$$W'_-(1; \underline{\rho}_A^{Nash}) < 0$$

implies that there exists $\varepsilon > 0$ such that

$$W'_-(1; \rho) < 0 \quad \forall \rho \in [\underline{\rho}_A^{Nash}, \underline{\rho}_A^{Nash} + \varepsilon).$$

For each such ρ , the point $\alpha = 1$ is not even a local maximizer of $\alpha \mapsto W(\alpha; \rho)$, because moving slightly to the left increases welfare. Therefore it cannot be a global maximizer, and hence

$$\mathcal{S}_{SB} \cap [\underline{\rho}_A^{Nash}, \underline{\rho}_A^{Nash} + \varepsilon) = \emptyset.$$

It follows that

$$\rho_A^{SB} \geq \underline{\rho}_A^{Nash} + \varepsilon > \underline{\rho}_A^{Nash}.$$

This proves

$$\rho_A^{SB} > \underline{\rho}_A^{Nash}.$$

□

Remark 3.7 (Operational Efficiency and the Frictionless Limit). *Theorem 3.5 identifies a robust first-best motive for positive AI exposure when $\phi > 0$. Holding aggregate trading sensitivity fixed, replacing a marginal amount of human exposure with AI exposure lowers physical execution costs at first order because $1/\sqrt{\rho_A} < 1/\sqrt{\rho_H}$. At the pure-human boundary, the associated loss in trading revenue is only second order, since the shared-noise term enters quadratically in C_A . This is why the pure-human allocation cannot be first-best once execution frictions are present.*

As $\phi \downarrow 0$, this operational-cost channel disappears. In that limit, the planner's trade-off is governed entirely by information aggregation and the common-noise externality, rather than by execution-cost savings.

Remark 3.8 (Economic Intuition of the $Q(0) = 1/2$ Threshold). *The threshold $Q(0) = 1/2$ should be interpreted as a boundary-capacity benchmark for the constrained planner. The expression for $W'(0)$ shows that the welfare effect of marginal AI entry at the origin is determined entirely by whether the baseline human market under- or over-harvests information.*

This connects the model to the classical Kyle benchmark (Kyle, 1985), in which profits are maximized when only half of the fundamental information is incorporated into prices. If $Q(0) < 1/2$, a marginal increase in AI adoption improves price discovery faster than it worsens trading synchronization. If $Q(0) > 1/2$, the market is already operating beyond that benchmark, and additional synchronization is welfare reducing at the origin.

Remark 3.9 (Why the monotone-crowding condition is needed). *The condition*

$$E(\alpha) \leq E(0) \quad \forall \alpha \in [0, 1]$$

is necessary. The decomposition

$$W'(\alpha) = \Delta(\alpha) + E(\alpha), \quad \Delta(\alpha) = U_A(\alpha) - U_H(\alpha), \quad E(\alpha) = \alpha U'_A(\alpha) + (1 - \alpha) U'_H(\alpha),$$

separates two forces. The term $\Delta(\alpha)$ is the private adoption gain of the marginal trader. Lemma 3.2 shows that this term is decreasing whenever AI adoption is privately attractive. The term $E(\alpha)$, by contrast, captures the effect of expanding AI adoption on the continuation payoffs of traders already in the market. Economically, it is the endogenous crowding component: it measures how the equilibrium trading environment changes as order flow becomes more synchronized.

The monotone-crowding condition says that this endogenous component does not improve as AI adoption expands. In an over-informative market, the pure-human benchmark already extracts too much information relative to the $Q(0) = 1/2$ capacity benchmark. If the crowding component is weakly decreasing, then additional AI adoption cannot create a later recovery in marginal surplus. The condition is therefore not a cosmetic technical assumption. It is the additional economic restriction that rules out a non-monotone recovery of second-best surplus at higher AI adoption rates.

Remark 3.10 (First-Best and Decentralized Thresholds). *Theorem 3.5 and Theorem 3.6 introduce conceptually distinct threshold objects. The first-best threshold ρ_A^{FB} is defined by the unconstrained planner's global problem after jointly optimizing over adoption and trading quantities. By contrast, the decentralized threshold ρ_A^{Nash} is determined by the private profitability of full AI adoption under endogenous price impact. These two thresholds arise from different optimization problems and therefore need not admit a universal algebraic ordering.*

4 Heterogeneous Adoption Costs

The analysis so far has isolated the crowding externality generated by a shared AI signal in an environment without explicit technology adoption costs. In that benchmark, AI adoption is governed entirely by the private continuation-value difference $\Delta(\alpha) = U_A(\alpha) - U_H(\alpha)$, and equilibrium adoption can naturally collapse to a boundary outcome. This stark structure is useful for exposing the core mechanism, but real AI adoption is rarely costless or homogeneous. Traders differ in their ability to access, integrate, validate, and operationalize AI-based signals. Some institutions can deploy such tools at low marginal cost, while others face compliance, infrastructure, or expertise constraints.

This section introduces heterogeneous private adoption costs. The purpose is twofold. First, it shows that the AI monoculture paradox is not merely an artifact of homogeneous traders or corner adoption. With heterogeneous costs, decentralized adoption is typically gradual and may be interior: only traders whose private adoption costs are below the equilibrium adoption gain choose the AI signal. Second, the cost distribution allows us to distinguish gross informed-sector surplus from net informed-sector surplus. Whereas $W(\alpha) = \alpha U_A(\alpha) + (1 - \alpha) U_H(\alpha)$ captures the trading surplus generated by the equilibrium allocation, the relevant net objective subtracts the total technology adoption cost borne by AI adopters.

The main insight is that heterogeneity smooths the adoption process but does not remove the externality. A marginal adopter compares only her private gain $\Delta(\alpha)$ with her private cost. The planner, by contrast, also accounts for how the additional AI adopter changes the trading environment for all other market participants through greater synchronization of order flow. Consequently, the decentralized market can still over-adopt AI relative to the adoption-only planner, and full AI adoption can become privately sustainable strictly before it is justified under the net informed-sector surplus criterion.

4.1 Model

Let the private cost of adopting the AI signal for trader i be

$$c_i \stackrel{i.i.d.}{\sim} F,$$

where F is a strictly increasing and absolutely continuous cumulative distribution function on $[0, \bar{c}]$, with continuous density f satisfying

$$f(c) > 0 \quad \forall c \in (0, \bar{c}).$$

A trader rationally adopts the AI signal if and only if the utility gain exceeds their

private cost:

$$c_i \leq U_A(\alpha) - U_H(\alpha) \equiv \Delta(\alpha).$$

To express aggregate welfare as a function of the macroscopic adoption rate α , note that the α -fraction of traders who adopt are precisely those with the lowest adoption costs. The marginal adopter therefore has cost

$$c^* = F^{-1}(\alpha).$$

Hence the total adoption cost borne by the adopting population is

$$C(\alpha) = \int_0^{F^{-1}(\alpha)} c f(c) dc.$$

By the Fundamental Theorem of Calculus and the strict monotonicity of F ,

$$C'(\alpha) = F^{-1}(\alpha).$$

We can now define net informed-sector surplus as

$$W^{net}(\alpha) = \underbrace{\alpha U_A(\alpha) + (1 - \alpha) U_H(\alpha)}_{\text{gross trading welfare } W(\alpha)} - C(\alpha).$$

4.2 Equilibrium

Lemma 4.1 (Existence and Uniqueness of Equilibrium). *Assume $\rho_A > \rho_H$ and let $\bar{c} := F^{-1}(1)$. Then the decentralized adoption game admits a unique equilibrium*

$$\alpha^* \in (0, 1]$$

solving

$$\alpha^* = F(\Delta(\alpha^*)).$$

More precisely:

$$\Delta(1) < \bar{c} \implies \alpha^* \in (0, 1),$$

whereas

$$\Delta(1) \geq \bar{c} \implies \alpha^* = 1.$$

Moreover, $\Delta(1)$ is strictly increasing in ρ_A . Hence, whenever

$$\bar{c} \leq \Delta(1) \Big|_{\rho_A=1},$$

there exists a unique decentralized full-adoption threshold

$$\tilde{\rho}_A^{Nash} \in (\rho_H, 1]$$

defined by

$$\Delta(1) \Big|_{\rho_A = \tilde{\rho}_A^{Nash}} = \bar{c}. \quad (25)$$

Proof. Define

$$G(\alpha) := F(\Delta(\alpha)) - \alpha.$$

Since $\rho_A > \rho_H$, we have $\Delta(0) > 0$. Let

$$\bar{\alpha} := \sup\{\alpha \in [0, 1] : \Delta(\alpha) > 0\}.$$

By continuity of Δ and Lemma 3.2, the set

$$A_+ := \{\alpha \in [0, 1] : \Delta(\alpha) > 0\}$$

is an initial interval. Indeed, if a second positive component existed after Δ had reached zero, then at the left endpoint of that component continuity would give $\Delta = 0$, while Lemma 3.2 would force Δ to be strictly decreasing immediately to the right, making it impossible for Δ to become positive there. Hence $A_+ = [0, \bar{\alpha})$ or $A_+ = [0, 1]$.

Therefore, since F is increasing and Δ is strictly decreasing on A_+ , the composition $F(\Delta(\alpha))$ is weakly decreasing on $[0, 1]$. It is equal to zero whenever $\Delta(\alpha) \leq 0$. Consequently, $G(\alpha) = F(\Delta(\alpha)) - \alpha$ is strictly decreasing on $[0, 1]$.

At $\alpha = 0$, since $\rho_A > \rho_H$, we have $\Delta(0) > 0$, hence

$$G(0) = F(\Delta(0)) > 0.$$

At $\alpha = 1$,

$$G(1) = F(\Delta(1)) - 1 \leq 0.$$

By the Intermediate Value Theorem, there exists a unique root

$$\alpha^* \in (0, 1]$$

such that

$$G(\alpha^*) = 0.$$

If $\Delta(1) < \bar{c}$, then $F(\Delta(1)) < 1$, so $G(1) < 0$, and the unique root must lie in $(0, 1)$. If $\Delta(1) \geq \bar{c}$, then $F(\Delta(1)) = 1$, so $G(1) = 0$, and uniqueness implies $\alpha^* = 1$.

It remains to show that $\Delta(1)$ is strictly increasing in ρ_A . Let $x_1 := \frac{\kappa(1)}{\phi}$ so that

$$x_1(1+x_1)^2 = \rho_A \frac{\sigma_v^2}{\phi^2 \sigma_\epsilon^2}.$$

At $\alpha = 1$,

$$\Delta(1) = \frac{\sigma_v^2}{2\phi(1+x_1)^2} \left[\rho_A - \rho_H(1 + (1-\rho_A)x_1)^2 \right].$$

Differentiating the implicit equation for x_1 gives

$$\frac{dx_1}{d\rho_A} = \frac{x_1(1+x_1)}{\rho_A(1+3x_1)}.$$

A direct differentiation then yields

$$\frac{d}{d\rho_A} \Delta(1) = \frac{\sigma_v^2}{2\phi(1+x_1)^2(1+3x_1)} \left[1 + x_1 + 4\rho_H x_1 + (10 - 4\rho_A)\rho_H x_1^2 + 6(1-\rho_A)\rho_H x_1^3 \right] > 0.$$

Therefore, $\Delta(1)$ is strictly increasing in ρ_A , and the threshold $\tilde{\rho}_A^{Nash}$ defined by (25) is unique whenever it exists. $\square$

Theorem 4.2 (Delayed Full Adoption with Heterogeneous Costs). *Assume the decentralized full-adoption threshold $\tilde{\rho}_A^{Nash}$ exists. Define the planner's boundary full-adoption threshold by*

$$\tilde{\rho}_A^{SB} := \inf \{ \rho_A \in (\rho_H, 1] : W^{net'}(1; \rho_A) \geq 0 \}, \quad (26)$$

provided the set is nonempty. Then

$$W_-^{net'}(1) \Big|_{\rho_A = \tilde{\rho}_A^{Nash}} < 0. \quad (27)$$

Consequently, if $\tilde{\rho}_A^{SB}$ exists, it satisfies

$$\tilde{\rho}_A^{Nash} < \tilde{\rho}_A^{SB}. \quad (28)$$

If the set in (26) is empty, then complete AI monoculture is never planner-admissible on the physical domain $\rho_A \in (\rho_H, 1]$, so the "delay" is even stronger.

Proof. Let

$$x_1 := \frac{\kappa(1)}{\phi}, \quad r := 1 - \rho_A.$$

At $\alpha = 1$, equations (20)–(21) imply

$$U_A(1) = \frac{\sigma_v^2 \rho_A}{2\phi(1+x_1)^2}, \quad U_H(1) = \frac{\sigma_v^2 \rho_H (1+rx_1)^2}{2\phi(1+x_1)^2}.$$

Hence

$$\Delta(1) = \frac{\sigma_v^2}{2\phi(1+x_1)^2} \left[\rho_A - \rho_H(1+rx_1)^2 \right].$$

At the decentralized full-adoption threshold $\rho_A = \tilde{\rho}_A^{Nash}$, we have

$$\Delta(1) = \bar{c} \geq 0,$$

so necessarily

$$\rho_H(1+rx_1)^2 \leq \rho_A. \quad (29)$$

Now differentiate the equilibrium system at $\alpha = 1$. A direct simplification yields

$$U'_A(1) = \frac{\sigma_v^2 x_1}{\phi(1+x_1)^2(1+3x_1)} \left[\rho_H((1+\rho_A) + r(2+\rho_A)x_1 + r^2 x_1^2) - 2\rho_A \right].$$

Using (29),

$$\rho_H \leq \frac{\rho_A}{(1+rx_1)^2},$$

hence

$$\begin{aligned} & \rho_H((1+\rho_A) + r(2+\rho_A)x_1 + r^2 x_1^2) - 2\rho_A \\ & \leq \rho_A \frac{(1+\rho_A) + r(2+\rho_A)x_1 + r^2 x_1^2}{(1+rx_1)^2} - 2\rho_A. \end{aligned}$$

But

$$2(1+rx_1)^2 - ((1+\rho_A) + r(2+\rho_A)x_1 + r^2 x_1^2) = r(1+x_1)(1+rx_1) > 0$$

whenever $\rho_A < 1$. Therefore the bracket is strictly negative. If $\rho_A = 1$, then the bracket reduces to

$$2\rho_H - 2 < 0.$$

Thus, in all cases,

$$U'_A(1) \Big|_{\rho_A = \tilde{\rho}_A^{Nash}} < 0.$$

Finally, at $\alpha = 1$,

$$W_-^{net'}(1) = \Delta(1) - \bar{c} + U'_A(1).$$

Therefore, at $\rho_A = \tilde{\rho}_A^{Nash}$,

$$W_-^{net'}(1) \Big|_{\rho_A = \tilde{\rho}_A^{Nash}} = \Delta(1) - \bar{c} + U'_A(1) = U'_A(1) < 0,$$

which proves (27).

Since $W_-^{net'}(1; \rho_A)$ is continuous in ρ_A , the strict negativity at $\rho_A = \tilde{\rho}_A^{Nash}$ implies that it remains negative on a right neighborhood of $\tilde{\rho}_A^{Nash}$. Hence $\tilde{\rho}_A^{SB} > \tilde{\rho}_A^{Nash}$. $\square$

4.3 Heterogeneous AI Monoculture Paradox

Marginal decomposition of net surplus. The heterogeneous-cost environment makes the adoption problem gradual rather than all-or-nothing. The key question is whether net informed-sector surplus can still increase after the decentralized market has reached its equilibrium adoption rate α^* . The answer depends on two distinct marginal forces.

First, the marginal switcher compares her private adoption gain with her private adoption cost. This component is

$$S(\alpha) := \Delta(\alpha) - F^{-1}(\alpha).$$

At the decentralized equilibrium,

$$S(\alpha^*) = 0,$$

because the marginal adopter is privately indifferent:

$$\Delta(\alpha^*) = F^{-1}(\alpha^*).$$

Moreover, since

$$G(\alpha) = F(\Delta(\alpha)) - \alpha$$

is strictly decreasing and $G(\alpha^*) = 0$, we have

$$\Delta(\alpha) \leq F^{-1}(\alpha) \quad \forall \alpha \in [\alpha^*, 1].$$

Thus,

$$S(\alpha) \leq 0 \quad \forall \alpha \in [\alpha^*, 1].$$

In words, once the decentralized market has reached α^* , any further adopter is no longer privately valuable net of her own adoption cost.

Second, an additional AI adopter changes the trading environment faced by all other market participants. This spillover is captured by

$$E(\alpha) := \alpha U'_A(\alpha) + (1 - \alpha) U'_H(\alpha).$$

Therefore,

$$W^{net'}(\alpha) = W'(\alpha) - F^{-1}(\alpha) = \underbrace{\Delta(\alpha) - F^{-1}(\alpha)}_{\text{marginal switcher's net private surplus}} + \underbrace{E(\alpha)}_{\text{spillover on incumbent traders}}.$$

The first term is pinned down by the decentralized fixed point and is nonpositive on the right tail $[\alpha^*, 1]$. The hard part is the second term: additional AI adoption may improve or worsen incumbent traders' continuation payoffs through changes in synchronization, price

impact, and equilibrium trading intensities. The following proposition gives a general diagnostic criterion, while the two corollaries provide economically interpretable conditions under which the diagnostic is satisfied.

Proposition 4.3 (Right-Tail Diagnostic for Over-Adoption). *Assume the decentralized equilibrium $\alpha^* \in (0, 1)$ is the unique fixed point characterized in Lemma 4.1. If*

$$W^{net'}(\alpha) \leq 0 \quad \forall \alpha \in [\alpha^*, 1],$$

with strict inequality at $\alpha = \alpha^$, then every second-best maximizer satisfies*

$$\alpha^{SB} < \alpha^*, \quad W^{net}(\alpha^*) < W^{net}(\alpha^{SB}).$$

Equivalently, using the marginal decomposition

$$W^{net'}(\alpha) = [\Delta(\alpha) - F^{-1}(\alpha)] + E(\alpha),$$

the condition can be written as

$$E(\alpha) \leq F^{-1}(\alpha) - \Delta(\alpha) \quad \forall \alpha \in [\alpha^*, 1],$$

with strict inequality at $\alpha = \alpha^$.*

Proof. If $W^{net'}(\alpha) \leq 0$ on $[\alpha^*, 1]$ and the inequality is strict at α^* , then W^{net} is already locally decreasing at α^* and never increases to the right. Hence no maximizer of W^{net} can lie in $[\alpha^*, 1]$. Since W^{net} is continuous on the compact interval $[0, 1]$, a second-best maximizer exists and must satisfy $\alpha^{SB} < \alpha^*$. The strict welfare comparison follows immediately from optimality of α^{SB} . $\square$

Corollary 4.4 (Cost-Tail Dominance). *Assume*

$$E(\alpha^*) < 0, \quad W''(\alpha) \leq (F^{-1})'(\alpha) = \frac{1}{f(F^{-1}(\alpha))} \quad \forall \alpha \in [\alpha^*, 1].$$

Then

$$\alpha^{SB} < \alpha^*, \quad W^{net}(\alpha^*) < W^{net}(\alpha^{SB}).$$

Proof. At the decentralized equilibrium,

$$W^{net'}(\alpha^*) = [\Delta(\alpha^*) - F^{-1}(\alpha^*)] + E(\alpha^*) = E(\alpha^*) < 0.$$

Moreover,

$$W^{net''}(\alpha) = W''(\alpha) - (F^{-1})'(\alpha) \leq 0 \quad \forall \alpha \in [\alpha^*, 1].$$

Thus $W^{net'}$ is weakly decreasing on $[\alpha^*, 1]$. Since it is already strictly negative at α^* , it remains strictly negative throughout the right tail. The result follows from Proposition 4.3. $\square$

Economic meaning of the tail-curvature condition. The sign restriction

$$E(\alpha^*) < 0$$

means that at the very point where the marginal trader is privately indifferent, her entry still imposes a negative congestion spillover on the rest of the market. This is the externality ignored by the decentralized equilibrium.

The curvature restriction

$$W''(\alpha) \leq \frac{1}{f(F^{-1}(\alpha))}$$

compares the curvature of gross trading welfare to the slope of the marginal adoption cost schedule. Because

$$(F^{-1})'(\alpha) = \frac{1}{f(F^{-1}(\alpha))},$$

a thin right tail of the cost distribution (small f near the upper quantiles) makes marginal adoption costs rise steeply. Economically, this means that once the low-cost adopters have already entered, attracting additional traders becomes increasingly expensive. Therefore, even if gross trading welfare still rises slightly on the right tail, net welfare can already be falling because the cost of recruiting the remaining high-cost adopters rises even faster. This provides a natural economic force pushing the planner to stop strictly before the decentralized market does.

Corollary 4.5 (Uniform Crowding-Spillover Dominance). *Define*

$$\lambda := \frac{\sigma_v^2}{\phi^2 \sigma_\epsilon^2}.$$

Assume

$$\lambda < \frac{\left(\frac{\rho_A}{\rho_H}\right)^{1/3} - 1}{\rho_A(1 - \rho_A)}. \quad (\text{MC})$$

Then

$$E(\alpha) < 0 \quad \forall \alpha \in (0, 1].$$

Consequently,

$$\alpha^{SB} < \alpha^*, \quad W^{net}(\alpha^*) < W^{net}(\alpha^{SB}).$$

Proof. Let

$$x(\alpha) := \frac{\kappa(\alpha)}{\phi}, \quad r := 1 - \rho_A, \quad A(\alpha) := 1 + \alpha r x(\alpha).$$

From the equilibrium system, one can show that

$$x(\alpha) \leq \lambda \rho_A \quad \forall \alpha \in [0, 1].$$

Hence

$$A(\alpha) = 1 + \alpha(1 - \rho_A)x(\alpha) \leq 1 + \lambda \rho_A(1 - \rho_A).$$

Since

$$1 + \lambda \rho_A(1 - \rho_A) < \left(\frac{\rho_A}{\rho_H} \right)^{1/3},$$

so

$$A(\alpha)^3 < \frac{\rho_A}{\rho_H} \quad \forall \alpha \in (0, 1].$$

This implies

$$U'_H(\alpha) < 0 \quad \forall \alpha \in (0, 1].$$

Moreover, the same inequality yields

$$U'_A(\alpha) \leq U'_H(\alpha) < 0 \quad \forall \alpha \in (0, 1].$$

Therefore,

$$E(\alpha) = \alpha U'_A(\alpha) + (1 - \alpha) U'_H(\alpha) < 0 \quad \forall \alpha \in (0, 1].$$

Moreover, by Lemma 4.1, the decentralized equilibrium α^* is unique. Since $G(\alpha) = F(\Delta(\alpha)) - \alpha$ is strictly decreasing and $G(\alpha^*) = 0$, we have

$$\Delta(\alpha) \leq F^{-1}(\alpha) \quad \forall \alpha \in [\alpha^*, 1].$$

Therefore, for every $\alpha \in [\alpha^*, 1]$,

$$W^{net'}(\alpha) = [\Delta(\alpha) - F^{-1}(\alpha)] + E(\alpha) < 0.$$

The conclusion follows from Proposition 4.3. □

Economic meaning of Assumption MC. This condition provides a primitive route to over-adoption by making the spillover term uniformly negative. Under Assumption MC, additional AI adoption reduces the continuation payoffs of both AI and human traders throughout the adoption region:

$$U'_A(\alpha) \leq U'_H(\alpha) < 0.$$

Therefore, the incumbent-spillover term

$$E(\alpha) = \alpha U'_A(\alpha) + (1 - \alpha) U'_H(\alpha)$$

is strictly negative for every $\alpha > 0$. Once the decentralized equilibrium has been reached, the marginal switcher's own net surplus is already nonpositive; with uniformly negative spillovers, net surplus must decline on the entire right tail.

Economically, Assumption MC rules out the possibility that expanding AI adoption improves the trading environment for incumbent traders. It is therefore a crowding-spillover dominance condition: the synchronization effect of additional AI adoption dominates any continuation-value improvement generated by the AI signal.

5 The Blind Spot: Systematic Bias and Model Misspecification

Sections 2–4 intentionally study the AI signal under a favorable benchmark: the shared AI error is common across adopters but structurally unbiased. The resulting paradox is therefore a second-moment phenomenon. A common AI signal synchronizes order flow and reduces the diversification benefit generated by idiosyncratic human errors, even when the AI signal is more precise.

We now show that the same monoculture logic becomes more severe when the common AI error has a first-moment component. Specifically, suppose that the true AI signal contains a deterministic systematic bias, while traders and market makers continue to behave as if the AI signal were unbiased. This extension is not a separate mechanism, but the first-moment analogue of the same monoculture force: in Sections 2–4, the common AI error raises the covariance of trades; here, a common AI bias shifts the mean of trades.

5.1 Subjective Beliefs and the Objective Data-Generating Process

Let

$$s^0 := v + \epsilon_{AI}$$

denote the unbiased AI signal studied in the baseline model. In this section, the true AI signal is instead

$$s^b := v + \epsilon_{AI} + b, \quad b \neq 0, \tag{30}$$

where b is a deterministic but unobserved systematic bias. We interpret b as a structural model error or algorithmic blind spot: the AI oracle is not merely noisy, but systematically shifted in one direction.

To formalize model misspecification, we distinguish two probability models. Under the subjective model $\mathbb{P}^0$, agents believe

$$s = s^0 = v + \epsilon_{AI}.$$

Under the objective model $\mathbb{P}^b$, the true signal is instead

$$s = s^b = v + \epsilon_{AI} + b.$$

The primitives $v, \epsilon_{AI}, \eta_i, \epsilon_P$ remain mutually independent, with the same variances as in the baseline model. The only change is that the AI signal contains the deterministic shift b .

Throughout this section, traders and market makers are assumed to operate under the misspecified belief $\mathbb{P}^0$. Equivalently, they are overconfident in the AI oracle and do not statistically learn or correct the bias within the trading horizon. This is appropriate for a one-shot trading environment, or for a setting in which the structural error is latent relative to the decision horizon. Thus the trading coefficients and pricing coefficient are still the Section 3 equilibrium objects

$$c_A(\alpha), \quad c_H(\alpha), \quad \kappa(\alpha),$$

computed as if $b = 0$.

For clarity, define the subjective continuation values

$$U_A^0(\alpha), \quad U_H^0(\alpha)$$

as the ex-ante expected profits evaluated under $\mathbb{P}^0$, using the misspecified equilibrium strategies. These are exactly the baseline objects from Section 3:

$$U_A^0(\alpha) = U_A(\alpha), \quad U_H^0(\alpha) = U_H(\alpha).$$

By contrast, the objective continuation values

$$U_A^b(\alpha), \quad U_H^b(\alpha)$$

are the true ex-ante expected profits under $\mathbb{P}^b$, holding fixed the same misspecified trading coefficients and pricing rule.

This distinction is central. Decentralized adoption is governed by the subjective private gain

$$\Delta^0(\alpha) := U_A^0(\alpha) - U_H^0(\alpha),$$

whereas the realized informed-sector surplus is governed by the objective payoffs $U_A^b(\alpha)$ and $U_H^b(\alpha)$.

5.2 Misspecified Equilibrium and the Bias Penalty

Fix an aggregate AI adoption rate $\alpha \in [0, 1]$. Under the subjective model, the aggregate order flow is the baseline order flow

$$y^0(\alpha) = \alpha c_A(\alpha)(v + \epsilon_{AI}) + (1 - \alpha)c_H(\alpha)v + \epsilon_P. \quad (31)$$

The market maker, operating under $\mathbb{P}^0$, uses the subjectively efficient linear pricing rule

$$P^0(y) = \kappa(\alpha)y. \quad (32)$$

Under the true model $\mathbb{P}^b$, however, AI adopters trade on $s^b = v + \epsilon_{AI} + b$. Hence the realized order flow becomes

$$y^b(\alpha) = \underbrace{\alpha c_A(\alpha)(v + \epsilon_{AI}) + (1 - \alpha)c_H(\alpha)v + \epsilon_P}_{\text{baseline order flow } y^0(\alpha)} + \underbrace{\alpha c_A(\alpha)b}_{\text{common directional shift}}. \quad (33)$$

Thus

$$y^b(\alpha) = y^0(\alpha) + \alpha c_A(\alpha)b.$$

It is important to emphasize that $P^0(y) = \kappa(\alpha)y$ is only subjectively semi-strong efficient. If market makers knew the bias b , then the efficient price under the objective model would instead be

$$\mathbb{E}^b[v \mid y^b] = \kappa(\alpha)(y^b - \alpha c_A(\alpha)b),$$

because the deterministic shift $\alpha c_A(\alpha)b$ carries no information about the fundamental value. Therefore, Section 5 studies a misspecified equilibrium: agents coordinate on the same trading and pricing rules as in Section 3, but those rules are evaluated under the true biased data-generating process.

The following lemma gives the exact objective payoff distortion induced by the common bias.

Lemma 5.1 (The Asymmetric Penalty of a Common Bias). *Fix $\alpha \in [0, 1]$. Suppose traders and market makers use the misspecified equilibrium coefficients $c_A(\alpha)$, $c_H(\alpha)$, $\kappa(\alpha)$ computed under $\mathbb{P}^0$, while the true AI signal is*

$$s^b = v + \epsilon_{AI} + b.$$

Then the true ex-ante expected payoff of a human-signal trader is unchanged:

$$U_H^b(\alpha) = U_H^0(\alpha) = U_H(\alpha). \quad (34)$$

By contrast, the true ex-ante expected payoff of an AI-signal trader is

$$U_A^b(\alpha) = U_A(\alpha) - \underbrace{c_A(\alpha)^2 b^2 \left(\kappa(\alpha) \alpha + \frac{\phi}{2} \right)}_{\text{systematic bias penalty}}.$$

Proof. Let

$$\tilde{s} := v + \epsilon_{AI}$$

denote the unbiased component of the AI signal, and write the baseline order flow as

$$\tilde{y} := \alpha c_A(\alpha) \tilde{s} + (1 - \alpha) c_H(\alpha) v + \epsilon_P.$$

Under the true model,

$$s^b = \tilde{s} + b, \quad y^b = \tilde{y} + \alpha c_A(\alpha) b.$$

For readability, suppress the dependence of c_A, c_H, κ on α .

First consider a human-signal trader. Her trading quantity is

$$q_H = c_H \theta_i, \quad \theta_i = v + \eta_i.$$

Her true ex-ante expected payoff is

$$U_H^b(\alpha) = \mathbb{E}^b \left[c_H \theta_i (v - \kappa(\tilde{y} + \alpha c_A b)) - \frac{\phi}{2} c_H^2 \theta_i^2 \right].$$

Expanding the deterministic shift gives

$$U_H^b(\alpha) = \mathbb{E}^b \left[c_H \theta_i (v - \kappa \tilde{y}) - \frac{\phi}{2} c_H^2 \theta_i^2 \right] - \kappa \alpha c_A b c_H \mathbb{E}^b[\theta_i].$$

Since $\mathbb{E}^b[\theta_i] = \mathbb{E}[v + \eta_i] = 0$, the second term vanishes. Moreover, the first term is exactly the baseline payoff evaluated under the zero-mean variables $v, \epsilon_{AI}, \eta_i, \epsilon_P$. Hence

$$U_H^b(\alpha) = U_H^0(\alpha).$$

Now consider an AI-signal trader. Her trading quantity is

$$q_A = c_A s^b = c_A(\tilde{s} + b).$$

Her true ex-ante expected payoff is

$$U_A^b(\alpha) = \mathbb{E}^b \left[c_A(\tilde{s} + b)(v - \kappa(\tilde{y} + \alpha c_A b)) - \frac{\phi}{2} c_A^2 (\tilde{s} + b)^2 \right].$$

Expanding around the unbiased payoff gives

$$\begin{aligned} U_A^b(\alpha) &= \mathbb{E}^b \left[c_A \tilde{s}(v - \kappa \tilde{y}) - \frac{\phi}{2} c_A^2 \tilde{s}^2 \right] \\ &\quad + c_A b \mathbb{E}^b[v - \kappa \tilde{y}] - \kappa \alpha c_A^2 b \mathbb{E}^b[\tilde{s}] - \kappa \alpha c_A^2 b^2 \\ &\quad - \phi c_A^2 b \mathbb{E}^b[\tilde{s}] - \frac{\phi}{2} c_A^2 b^2. \end{aligned}$$

Because

$$\mathbb{E}^b[v] = 0, \quad \mathbb{E}^b[\tilde{y}] = 0, \quad \mathbb{E}^b[\tilde{s}] = 0,$$

all linear cross terms vanish. The first expectation is precisely $U_A^0(\alpha)$. Therefore

$$U_A^b(\alpha) = U_A^0(\alpha) - \kappa \alpha c_A^2 b^2 - \frac{\phi}{2} c_A^2 b^2,$$

which is

$$U_A^b(\alpha) = U_A^0(\alpha) - c_A(\alpha)^2 b^2 \left(\kappa(\alpha) \alpha + \frac{\phi}{2} \right).$$

This proves the claim. $\square$

Lemma 5.1 shows that systematic bias affects the two signal types asymmetrically. Human-signal traders have unchanged ex-ante expected payoff because their individual signals remain mean-zero and their aggregate idiosyncratic errors still cancel in the continuum. This does not mean that their realized payoff distribution is unaffected; rather, their ex-ante expectation is unchanged.

AI-signal traders, by contrast, all condition on the same shifted signal. The common bias therefore survives aggregation and generates a deterministic quadratic payoff loss. The penalty has two components. The term

$$\frac{\phi}{2} c_A(\alpha)^2 b^2$$

is the physical execution cost of trading on the biased component of the signal. The term

$$\kappa(\alpha) \alpha c_A(\alpha)^2 b^2$$

is the price-impact component: when many traders act on the same biased signal, their collective order flow shifts prices against themselves. **Thus the blind spot is not merely an individual forecasting error; it is amplified by monoculture.**

5.3 The Vulnerable AI Monoculture Paradox

We now use the subjective–objective distinction introduced above to state the biased version of the AI monoculture paradox. The key point is that model misspecification separates decentralized adoption incentives from realized informed-sector surplus. Traders choose whether to adopt the AI signal according to their subjective continuation values, whereas welfare is evaluated under the true data-generating process.

Because traders and market makers behave under the subjective model $\mathbb{P}^0$, their perceived continuation values are identical to those in the unbiased benchmark:

$$U_A^0(\alpha) = U_A(\alpha), \quad U_H^0(\alpha) = U_H(\alpha).$$

Consequently, the subjective private adoption gain is

$$\Delta^0(\alpha) := U_A^0(\alpha) - U_H^0(\alpha) = U_A(\alpha) - U_H(\alpha) = \Delta(\alpha).$$

Thus the decentralized full-adoption threshold ρ_A^{Nash} is unchanged from Theorem 3.3. In particular, when $\rho_A > \rho_A^{\text{Nash}}$, the subjective adoption game leads traders to coordinate on the full AI monoculture $\alpha^* = 1$. The systematic bias provides no endogenous deterrent at the adoption stage, because it is not priced into individual adoption incentives.

However, the objective informed-sector surplus is evaluated under the true model $\mathbb{P}^b$. Since the pure-human benchmark contains no AI signal, it is unchanged by the bias:

$$W^b(0) = U_H^b(0) = U_H^0(0) = U_H(0).$$

Under full AI adoption,

$$W^b(1) = U_A^b(1).$$

The biased monoculture paradox therefore occurs when traders subjectively enter the AI monoculture, yet the true ex-ante informed-sector surplus under full AI adoption is below the pure-human benchmark:

$$U_A^b(1) < U_H(0).$$

Theorem 5.2 (Biased AI Monoculture). *Consider the misspecified AI signal*

$$s^{AI} = v + \varepsilon_{AI} + b, \quad b \neq 0,$$

and suppose traders and market makers behave under the subjective model $b = 0$. Then the subjective adoption incentives are unchanged from Section 3. In particular, the decentralized full-adoption threshold ρ_A^{Nash} from Theorem 3.3 remains unchanged.

Define the full-adoption bias-amplification coefficient

$$\Gamma_b(\rho_A) := c_A(1; \rho_A)^2 \left(\kappa(1; \rho_A) + \frac{\phi}{2} \right).$$

Then $\Gamma_b(\rho_A) > 0$, and under full AI adoption the objective ex-ante payoff of an AI adopter is

$$U_A^b(1; \rho_A) = U_A(1; \rho_A) - b^2 \Gamma_b(\rho_A). \quad (35)$$

Hence full AI monoculture is objectively welfare-dominated relative to the pure-human benchmark if and only if

$$U_A(1; \rho_A) - b^2 \Gamma_b(\rho_A) < U_H(0). \quad (36)$$

Equivalently, whenever $U_A(1; \rho_A) > U_H(0)$, condition (36) can be written as

$$b^2 > \frac{U_A(1; \rho_A) - U_H(0)}{c_A(1; \rho_A)^2 \left(\kappa(1; \rho_A) + \frac{\phi}{2} \right)}. \quad (37)$$

Proof. Under the subjective model $\mathbb{P}^0$, traders and market makers believe that $b = 0$. Therefore the trading coefficients $c_A(\alpha), c_H(\alpha)$, the pricing coefficient $\kappa(\alpha)$, and the subjective continuation values are exactly the objects from the unbiased benchmark in Section 3. Hence the subjective adoption condition is also unchanged. In particular, the decentralized full-adoption threshold remains the threshold ρ_A^{Nash} characterized in Theorem 3.3.

It remains to evaluate the true payoff under $\mathbb{P}^b$. By Lemma 5.1, for any adoption rate α ,

$$U_A^b(\alpha) = U_A^0(\alpha) - c_A(\alpha)^2 b^2 \left(\kappa(\alpha) \alpha + \frac{\phi}{2} \right).$$

Since $U_A^0(\alpha) = U_A(\alpha)$, evaluating this identity at $\alpha = 1$ gives

$$U_A^b(1; \rho_A) = U_A(1; \rho_A) - c_A(1; \rho_A)^2 b^2 \left(\kappa(1; \rho_A) + \frac{\phi}{2} \right),$$

which is (35).

The welfare comparison follows because the pure-human benchmark is unaffected by the AI bias:

$$W^b(0) = U_H(0), \quad W^b(1) = U_A^b(1).$$

Therefore full AI monoculture is dominated by the pure-human benchmark if and only if

$$U_A^b(1; \rho_A) < U_H(0),$$

which is exactly (36). If $U_A(1; \rho_A) > U_H(0)$, rearranging this inequality gives (37). $\square$

Theorem 5.2 clarifies how systematic bias changes the baseline monoculture paradox. Since traders evaluate adoption under the subjective model $b = 0$, the decentralized adoption threshold is unchanged. By contrast, objective full-adoption surplus is strictly reduced:

$$U_A^b(1; \rho_A) = U_A(1; \rho_A) - b^2 \Gamma_b(\rho_A), \quad \Gamma_b(\rho_A) := c_A(1; \rho_A)^2 \left(\kappa(1; \rho_A) + \frac{\phi}{2} \right) > 0.$$

Therefore, relative to the unbiased benchmark, systematic bias weakly expands the set of precisions for which full AI adoption is welfare-dominated. Indeed, if

$$\mathcal{D}_0 := \{ \rho_A \in [\rho_A^{\text{Nash}}, 1] : U_A(1; \rho_A) < U_H(0) \}$$

denotes the welfare-dominated full-adoption region in the unbiased model, and

$$\mathcal{D}_b := \{ \rho_A \in [\rho_A^{\text{Nash}}, 1] : U_A^b(1; \rho_A) < U_H(0) \},$$

then

$$\mathcal{D}_0 \subseteq \mathcal{D}_b,$$

because $U_A^b(1; \rho_A) < U_A(1; \rho_A)$ for every $b \neq 0$. In this precise sense, hidden bias amplifies the welfare consequences of AI monoculture: the same subjective adoption incentives now lead to a strictly lower objective surplus.

The penalty $b^2 \Gamma_b(\rho_A)$ has two components. The execution-cost component

$$\frac{\phi}{2} c_A(1; \rho_A)^2 b^2$$

is the direct cost of trading on the biased part of the signal. The price-impact component

$$\kappa(1; \rho_A) c_A(1; \rho_A)^2 b^2$$

captures the fact that, under full AI adoption, all traders act on the same directional mistake and thereby move prices against themselves. Thus systematic bias does not replace the monoculture mechanism; it adds a first-moment distortion on top of the original second-moment crowding externality.

Remark 5.3 (Recovery sets and the boundary of the paradox). *In the unbiased benchmark of Section 3, the full-AI payoff $U_A(1; \rho_A)$ is monotone in ρ_A , which permits a clean recovery root ρ^* . Under systematic bias, the relevant objective payoff is instead*

$$U_A^b(1; \rho_A) = U_A(1; \rho_A) - b^2 \Gamma_b(\rho_A).$$

This expression need not be globally increasing in ρ_A . Higher perceived AI precision raises

the informational value of the signal, but it may also increase the misspecified trading intensity and thereby amplify the quadratic bias penalty.

For this reason, we do not define a unique biased recovery threshold without an additional monotonicity condition. Instead, define the biased recovery set

$$\mathcal{R}_b := \{\rho_A \in (\rho_H, 1] : U_A^b(1; \rho_A) \geq U_H(0)\}.$$

Only if $\mathcal{R}_b$ is known to be an upper interval is it natural to define a biased recovery threshold

$$\rho_b^* := \inf \mathcal{R}_b.$$

5.4 Planner Implications under Systematic Bias

We now revisit the planner's problem under systematic misspecification. The key distinction from Section 3 is that decentralized traders and market makers still behave according to the subjective unbiased model, whereas the planner evaluates objective ex-ante informed-sector surplus under the true data-generating process.

This distinction matters most for the adoption-only, second-best planner. If an unconstrained first-best planner observes the deterministic bias b and can implement bias-corrected trades based on $s^{AI} - b$, then the systematic bias is neutralized and the first-best problem reduces to the unbiased benchmark of Section 3. By contrast, the adoption-only planner cannot alter decentralized trading rules or the market maker's misspecified pricing rule. It can only choose the macroscopic adoption rate α . Hence the relevant second-best objective under systematic bias is

$$\alpha^{SB,b} \in \arg \max_{\alpha \in [0,1]} W^b(\alpha),$$

where

$$W^b(\alpha) := \alpha U_A^b(\alpha) + (1 - \alpha) U_H^b(\alpha).$$

By Lemma 5.1,

$$U_A^b(\alpha) = U_A(\alpha) - c_A(\alpha)^2 b^2 \left(\kappa(\alpha) \alpha + \frac{\phi}{2} \right), \quad U_H^b(\alpha) = U_H(\alpha).$$

Therefore

$$W^b(\alpha) = W(\alpha) - \mathcal{P}_b(\alpha), \tag{38}$$

where

$$\mathcal{P}_b(\alpha) := \alpha c_A(\alpha)^2 b^2 \left(\kappa(\alpha) \alpha + \frac{\phi}{2} \right). \tag{39}$$

The function $\mathcal{P}_b(\alpha)$ is the aggregate bias penalty borne by the AI-adopting population.

It is zero at $\alpha = 0$, but it affects the marginal planner incentives at both boundaries.

Proposition 5.4 (Second-best boundary effects of systematic bias). *Let $W^b(\alpha) = W(\alpha) - \mathcal{P}_b(\alpha)$, where $\mathcal{P}_b$ is defined in (39). Boundary derivatives are understood as one-sided derivatives.*

1. *At the no-AI boundary,*

$$(W^b)'_+(0) = W'_+(0) - \frac{\rho_A^2(1 - Q_0)^2}{2\phi} b^2, \quad (40)$$

where $Q_0 := Q(0)$. Equivalently, using the unbiased boundary formula from Section 3,

$$(W^b)'_+(0) = \frac{(1 - Q_0)^2}{2\phi} \left[\sigma_v^2(\rho_A - \rho_H) \left(\frac{1 - 2Q_0}{1 + 2Q_0} \right) - \rho_A^2 b^2 \right]. \quad (41)$$

Hence systematic bias strictly lowers the marginal value of initial AI adoption. In particular, if $Q_0 \geq 1/2$, then

$$(W^b)'_+(0) < 0$$

for every $b \neq 0$. If $Q_0 < 1/2$, then initial AI entry is locally second-best beneficial if and only if

$$b^2 < \frac{\sigma_v^2(\rho_A - \rho_H)}{\rho_A^2} \left(\frac{1 - 2Q_0}{1 + 2Q_0} \right).$$

2. *At the full-AI boundary,*

$$(W^b)'_-(1) = W'_-(1) - (\mathcal{P}_b)'_-(1), \quad (42)$$

where

$$(\mathcal{P}_b)'_-(1) = b^2 \frac{d}{d\alpha} \left[\alpha c_A(\alpha)^2 \left(\kappa(\alpha)\alpha + \frac{\phi}{2} \right) \right] \Big|_{\alpha=1-}. \quad (43)$$

Therefore, whenever $(\mathcal{P}_b)'_-(1) \geq 0$, systematic bias makes full AI adoption weakly harder to justify at the boundary:

$$(W^b)'_-(1) \leq W'_-(1).$$

In particular, if

$$W'_-(1) - (\mathcal{P}_b)'_-(1) < 0,$$

then moving slightly left from $\alpha = 1$ strictly raises objective surplus, so full AI adoption cannot be a local, and hence cannot be a global, second-best optimum.

Proof. For part (1), recall that

$$\mathcal{P}_b(\alpha) = \alpha c_A(\alpha)^2 b^2 \left(\kappa(\alpha)\alpha + \frac{\phi}{2} \right).$$

Since $c_A(\alpha)$ and $\kappa(\alpha)$ are continuous at $\alpha = 0$, we have

$$(\mathcal{P}_b)'_+(0) = c_A(0)^2 b^2 \cdot \frac{\phi}{2}.$$

Using the equilibrium coefficient at the no-AI boundary,

$$c_A(0) = \frac{\rho_A(1 - Q_0)}{\phi},$$

we obtain

$$(\mathcal{P}_b)'_+(0) = \frac{\rho_A^2(1 - Q_0)^2}{2\phi} b^2.$$

Thus

$$(W^b)'_+(0) = W'_+(0) - \frac{\rho_A^2(1 - Q_0)^2}{2\phi} b^2,$$

which proves (40).

To obtain the explicit expression, substitute the unbiased boundary derivative from Section 3:

$$W'_+(0) = U_H(0) \frac{\rho_A - \rho_H}{\rho_H} \left(\frac{1 - 2Q_0}{1 + 2Q_0} \right),$$

together with

$$U_H(0) = \frac{\sigma_v^2 \rho_H}{2\phi} (1 - Q_0)^2.$$

This yields

$$W'_+(0) = \frac{\sigma_v^2(\rho_A - \rho_H)(1 - Q_0)^2}{2\phi} \left(\frac{1 - 2Q_0}{1 + 2Q_0} \right).$$

Subtracting $(\mathcal{P}_b)'_+(0)$ gives (41). The sign implications follow immediately.

For part (2), equation (38) gives

$$W^b(\alpha) = W(\alpha) - \mathcal{P}_b(\alpha).$$

Taking the left derivative at $\alpha = 1$ yields

$$(W^b)'_-(1) = W'_-(1) - (\mathcal{P}_b)'_-(1),$$

with $(\mathcal{P}_b)'_-(1)$ given by (43). If

$$W'_-(1) - (\mathcal{P}_b)'_-(1) < 0,$$

then for $\alpha < 1$ sufficiently close to 1, the one-sided derivative condition implies

$$W^b(\alpha) > W^b(1).$$

Thus $\alpha = 1$ cannot be a local maximizer of the second-best objective, and therefore cannot be a global maximizer. $\square$

Remark 5.5 (Economic interpretation). *Proposition 5.4 shows how systematic misspecification modifies the second-best boundary logic from Section 3. In the unbiased model, the sign of $W'_+(0)$ is governed by the market-capacity threshold $Q_0 = 1/2$: if the pure-human market under-extracts information ($Q_0 < 1/2$), marginal AI entry can be beneficial; if it over-extracts information ($Q_0 > 1/2$), marginal AI entry is harmful.*

With systematic bias, this boundary condition becomes stricter. Equation (41) shows that the bias subtracts the positive term

$$\frac{\rho_A^2(1 - Q_0)^2}{2\phi}b^2$$

from the marginal value of initial AI adoption. Hence even in an under-informative market with $Q_0 < 1/2$, initial AI entry is justified only if the systematic bias is sufficiently small. In an over-informative market $Q_0 \geq 1/2$, systematic bias reinforces the no-AI prescription.

At the full-adoption boundary, the same principle applies locally. If the bias penalty is weakly increasing as the market approaches full AI adoption, then it lowers the left boundary derivative and makes complete AI monoculture harder to justify. Thus systematic bias does not change decentralized coordination incentives, but it makes the second-best planner more cautious at both boundaries.

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